HAW HAMBURG

BACHELOR'S THESIS

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TODO: English title

FAKULTÄT ELEKTRO-, MEDIEN- UND INFORMATIONSTECHNIK

Faculty of Electrical Engineering, Media and Information Technology

HOCHSCHULE FÜR ANGEWANDTE WISSENSCHAFTEN HAMBURG

Hamburg University of Applied Sciences

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TODO: English title

Bachelor's thesis based on the examination and study regulations of the degree program *Bachelor of Science Information Engineering*
at the Faculty of Electrical Engineering, Media and Information Technology of the University of Applied Sciences Hamburg

Supervising examiner: Prof. Dr.-Ing. Marc Hensel
Second examiner: TODO: Zweitgutachter

Submitted on: TODO: Einreichungsdatum

Josia Vickers

Title of Thesis
TODO: English title

Keywords
TODO: English keywords

Abstract
TODO: English abstract

Josia Vickers

Thema der Arbeit
TODO: deutscher Titel

Stichwörter
TODO: deutsche Schlagworte

Kurzzusammenfassung
TODO: deutsche Kurzfassung

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1 Introduction

This chapter describes the overall context of the thesis work, why nonlinearity is detrimental to navigational radar systems, and how digital predistortion is useful for combating these effects. The problem environment is then defined, and objectives of this thesis are established.

1.1 Motivation

Marine radar is the primary sensor for situational awareness and collision avoidance at sea, particularly at night, in fog, and in heavy precipitation, when optical observation of the surroundings is unreliable or impossible. To obtain both long detection range and fine range resolution without resorting to very high peak transmit power, the SSR-X radar employs pulse compression: a long, modulated pulse is transmitted and compressed in the receiver by matched filtering, so that the achievable range resolution is set by the signal bandwidth rather than by the pulse duration. The detection performance of such a system depends directly on the energy that the transmit chain can put on the target, and thus on the power amplifier at the end of that chain.

This is where a fundamental limitation arises. A power amplifier behaves approximately linearly only well below its saturation point; as the input drive increases, gain compression and phase distortion set in, described by the amplifier's AM-AM and Amplitude-to-phase (AM-PM) characteristics. The consequences are twofold. Spectrally, the nonlinearity produces intermodulation products and harmonics that broaden the transmitted spectrum. The International Standard IEC 62388 requires X-band maritime radar to transmit within the allocated band of 9.2–9.5 GHz [12, Sec. 6.2.1.1], so such spectral regrowth places compliance at risk and may interfere with neighbouring services and other radars. In the signal domain, the distorted transmit waveform no longer matches the receiver's compression filter. The compressed pulse loses peak amplitude and, more critically, its range sidelobes rise. Elevated sidelobes around a strong

target mask weaker targets in its immediate vicinity – a small vessel, a navigation buoy, or floating debris alongside a large ship – and, once the sidelobe level exceeds the detection threshold, they can appear as false targets on the radar image. Both outcomes directly undermine the reliability of the instrument the crew is expected to trust.

The conventional remedy is to operate the amplifier with a generous power back-off, keeping it inside its linear region. This is a costly compromise: the available output power, and with it the signal-to-noise ratio (SNR) at the receiver, is deliberately left unused, and the amplifier is run at poor power efficiency. Digital predistortion offers an alternative. By applying the inverse of the amplifier's nonlinear characteristic to the signal in the digital domain before amplification, the effective transfer characteristic of the amplifier chain is linearised, allowing the amplifier to be driven closer to saturation while remaining spectrally compliant. The gain is a purely digital one and requires no additional expensive analogue hardware.

If this work produces successful results, the benefit is therefore concrete. Additional SNR headroom translates into a longer detection range, or into the detection of targets with a smaller radar cross section at the same range, on hardware that already exists in the field. Reduced range sidelobes restore the ability to resolve weak targets close to strong ones and reduce the likelihood of false targets appearing on the display. For the radar developers, the work supplies a quantified understanding of how amplifier nonlinearity degrades their system and a validated means of correcting it; for the crews who navigate by the resulting radar image, it means a more trustworthy picture of their surroundings and a better chance of avoiding a collision. Thus, the work aims to improve the fidelity of the safety-critical radar navigation system without a significant rise in cost.

1.2 Environment

This thesis is carried out externally at NACOS Marine GmbH, a supplier of integrated marine navigation and radar systems, within the Radar Development Team – responsible for the development and maintenance of the company's radar transceivers amongst other devices. Amongst the developed devices is the SSR-X radar transceiver. This device, more specifically its pulse compression signal chain, serves as the subject of the work. In the transmit chain, the medium and long baseband pulses used in the radar's standard operating mode are generated in firmware, windowed, upconverted to

IF, converted to the analogue domain, and finally upconverted to RF before being amplified for transmission. The echo returns attenuated and delayed and passes through the receive chain, where it is amplified, downconverted to IF and to baseband, and filtered, before pulse compression is performed to maximise SNR and to determine the target range from the echo delay. The power amplifier chain of the transmit path and its effect on the compressed output form the focus of this thesis. This is described in more detail in Chapter 3.

The SSR-X is an existing, productive platform that is actively maintained and further developed; this work does not introduce a new radar but investigates and improves a well-defined part of an established one. Its results feed into the wider effort at NACOS Marine to increase radar performance and to safeguard spectral compliance with IEC 62388. The assessment of its effect on spectral and pulse compression metrics, is a central part of this thesis.

1.3 Objectives

The superordinate goal to which this thesis contributes is to increase the detection performance of the SSR-X pulse compression radar and to secure its spectral compliance without modifying its analogue hardware, by exploiting the amplifier headroom that nonlinearity currently makes unusable. Within this frame, the aim of this thesis is twofold: first, to understand the influence of power amplifier nonlinearity on the power amplifier chain of the SSR-X pulse compression radar, and second, to counteract these effects using digital predistortion. This aim is pursued by first developing a generic simulation to model the rough behaviour of the SSR-X pulse compression radar in software. Subsequently, the simulation will be finetuned through lab measurements on the real hardware to provide a precise model of the system and its behaviour. Finally, this simulated model will lay the groundwork for experiments on examining power amplifier nonlinearity effects and testing predistorton algorithms to counteract these. Results of these observations and experiments will be evaluated to draw definitive conclusions to enhance performance.

2 Fundamentals

This chapter covers the theoretical foundations and current state of technology essential for understanding this thesis work. The first section provides a basic introduction on what pulse compression radar is and how it generally works. The following two sections discuss the effects of nonlinearity and existing models that describe the nonlinearity exhibited by power amplifiers. Finally, predistortion is discussed - namely the general concept, how it can compensate for nonlinearity effects, and prominent predistortion algorithms in practice.

The goal of this chapter is not to cover the entire field of radar theory and nonlinearity at large, but rather to cover the key concepts and theory directly pertinent to this thesis work. This provides the reader context and prior knowledge to comprehend the work.

2.1 Pulse Compression Radar

A pulsed radar determines the range of a target by transmitting a pulse and measuring the round-trip delay τ_d of the returned echo [27]. A simplified block diagram of a

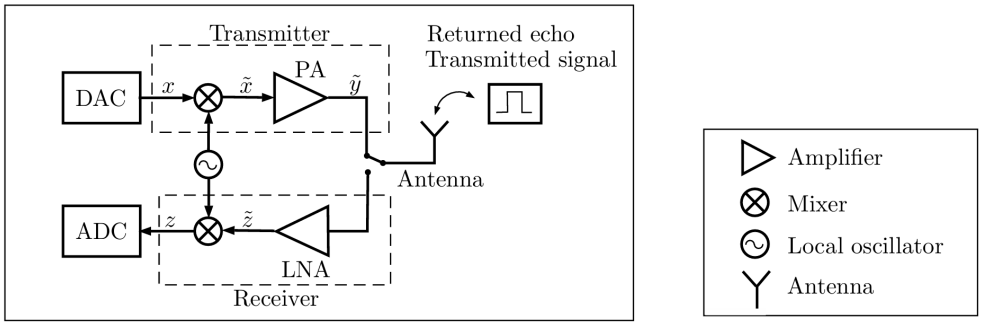


Figure 2.1: Simplified block diagram of pulsed radar system with explanation of symbols [10]

radar system is depicted in Figure 2.1. An analogue baseband signal x is output by the digital-to-analogue converter (DAC). The signal is subsequently upconverted to a carrier frequency by a mixer which multiplies the baseband signal with a sinusoidal signal generated by a local oscillator. The result is the RF signal $\tilde{x}$, then amplified by the power amplifier for transmission – yielding the amplified RF signal $\tilde{y}$. Following amplification, the antenna propagates the signal in space, and if a target is present, the antenna receives the energy of a returned echo. The received signal is often weak, and thus amplified before any processing or decisions regarding detection. As the detection is noise sensitive, the amplification is handled via a LNA, whose operation adds minimal noise. Before any processing and decisions whether a target is present, the signal $\tilde{z}$ is downconverted via mixing with the local oscillator and passed through an analogue-to-digital converter (ADC) – yielding the digital baseband signal z [36].

As the pulse wave propagates at the speed of light c in both directions (transmission and reception), the target range follows as

$$R = \frac{c\,\tau_d}{2}. \tag{2.1}$$

Two properties govern how well this is done. The first is detection performance, which is driven by the SNR at the receiver and therefore by the energy contained in the transmitted pulse – the product of its peak power and its duration [30]. The second is range resolution, the ability to separate two targets closely spaced in range. For a simple, unmodulated pulse of duration τ, two echoes are resolved only if they are separated by at least

$$\delta R = \frac{c\,\tau}{2}, \tag{2.2}$$

so that resolution improves as the pulse is shortened [30].

Equations (2.2) and the energy requirement are in direct conflict for an unmodulated pulse, as both are coupled through the pulse duration τ [30]. Lengthening the pulse raises the transmitted energy and thus the detection range, but degrades range resolution. Shortening the pulse improves resolution, but reduces the energy – which could only be recovered by raising the peak power, and peak power is limited by what the transmitter hardware can deliver [19]. Pulse compression resolves this conflict. By

modulating the signal within a long pulse, the waveform bandwidth B is increased independently of the pulse duration, such that $B \gg 1/\tau$ [30]. Range resolution is then set by the bandwidth rather than the pulse duration,

$$\delta R = \frac{c}{2B}, \tag{2.3}$$

and a long, low peak power pulse can therefore deliver the resolution of a short one.

2.1.1 Linear Frequency Modulation

The most widely used pulse compression waveform is the linear frequency modulated (LFM) waveform, also known as a chirp, in which the instantaneous frequency is swept linearly across the pulse [30]. A baseband LFM pulse is defined as

$$x(t) = A\cos\left(\pi\frac{B}{\tau}t^2\right), \qquad -\frac{\tau}{2} \leq t \leq \frac{\tau}{2}, \tag{2.4}$$

where A is the waveform amplitude, B the swept bandwidth and τ the pulse duration [30]. Differentiating the instantaneous phase gives the instantaneous frequency

$$f(t) = f_0 + k\,t, \qquad k = \frac{B}{\tau}, \tag{2.5}$$

where f_0 is the carrier frequency and k the frequency sweep rate [19]. The linear relationship of (2.5) is what gives the modulation its name. The sweep may be applied in either direction, producing an up-chirp or a down-chirp.

2.1.2 Matched Filtering and Pulse Compression

The received echo is corrupted by additive noise, and the quantity that governs the probability of detection is the output SNR rather than the shape of the received signal [19]. The filter that maximises this output SNR is the matched filter, whose impulse response is the complex conjugate of the time-reversed transmitted waveform [30]. Applying it is equivalent to correlating the received echo against a replica of what was transmitted, and its output is the convolution

$$y(t) = \int_{-\infty}^{\infty} h(\alpha)\, h^*(t-\alpha)\, \mathrm{d}\alpha, \tag{2.6}$$

where $h(t)$ is the impulse response of the waveform and $(\cdot)^*$ denotes complex conjugation [19]. For the LFM waveform of (2.4), evaluating (2.6) yields the compressed response

$$y(t) = \left(1 - \frac{|t|}{\tau}\right) \frac{\sin\left[\left(1 - \frac{|t|}{\tau}\right)\pi B t\right]}{\left(1 - \frac{|t|}{\tau}\right)\pi B t}, \qquad |t| \leq \tau, \tag{2.7}$$

which is the product of a triangular envelope and a term resembling a sinc function [30]. For time-bandwidth products greater than roughly 20 the response closely approximates a sinc, whose first null lies at $t \approx \pm 1/B$ – recovering the range resolution of (2.3).

The effect is therefore to concentrate the energy spread over the long transmitted pulse into a narrow, high-amplitude peak at the filter output. The degree of this concentration is quantified by the pulse compression ratio, which for an LFM waveform equals the time-bandwidth product

$$\mathrm{PCR} = \frac{\tau}{1/B} = \tau B. \tag{2.8}$$

The same product defines the pulse compression gain, the ratio of the SNR at the output of the matched filter to that at its input [30]. In modern systems, time-bandwidth products range from unity for a simple pulse to 10^6 and beyond. It is common for the SNR at the receiver input to lie below 0 dB, with the signal only rising above the noise floor once the matched filter has been applied [30].

2.1.3 Range Sidelobes

Applying the matched filter to a modulated waveform produces a response consisting of a mainlobe and an accompanying sidelobe structure [30]. The mainlobe is the portion of the response between the nulls adjacent to the peak, and its width determines the range resolution. The sidelobes, by contrast, are an undesired by-product of the modulation. They are problematic because the sidelobes of a strong target may exceed the mainlobe

of a weaker one, masking its presence even when the two targets are well resolved in range [30].

Sidelobe performance is quantified by two metrics. The peak side lobe level (PSL) is the ratio of the highest sidelobe to the peak of the mainlobe, and is approximately $-13.2\,\mathrm{dB}$ for an unweighted LFM waveform with a time-bandwidth product greater than 20 [30]. The integrated side lobe level (ISL) is the ratio of the total energy contained in the sidelobes to that in the mainlobe, and is approximately $-9.6\,\mathrm{dB}$ for the same waveform [30]. The latter is the more relevant metric in a distributed clutter environment, where the cumulative sidelobe contributions of many resolved clutter patches add noncoherently at the range of a target.

Sidelobes are commonly suppressed by applying an amplitude weighting, such as a Taylor window, to taper the transition regions of the waveform spectrum [30]. This constitutes a deliberate mismatch between the filter and the transmitted waveform, and consequently lowers the sidelobes at the cost of a broadened mainlobe – and therefore degraded resolution – along with a loss in processing gain.

Both the PSL and the ISL of the compressed response are used as key performance indicators (KPIs) in this thesis, as power amplifier nonlinearity distorts the transmitted waveform and consequently degrades the compressed response. The mechanisms by which this occurs are the subject of Section 2.2.

2.2 Effects of Nonlinearity

This section begins with a brief introduction to several concepts and equations paramount to understand nonlinearity and its effects. Firstly, a system can be called "memoryless" or "static" if its output depends solely on its current input value, and not on past input (or even output) values. Moreover, a memoryless system that is linear can be described by [29]

$$y(t) = \alpha\, x(t), \tag{2.9}$$

where $y(t)$ is the output signal, $x(t)$ the input signal and α the constant gain of the system. Equation (2.9) describes a linear relationship, as output is directly proportional to the input.

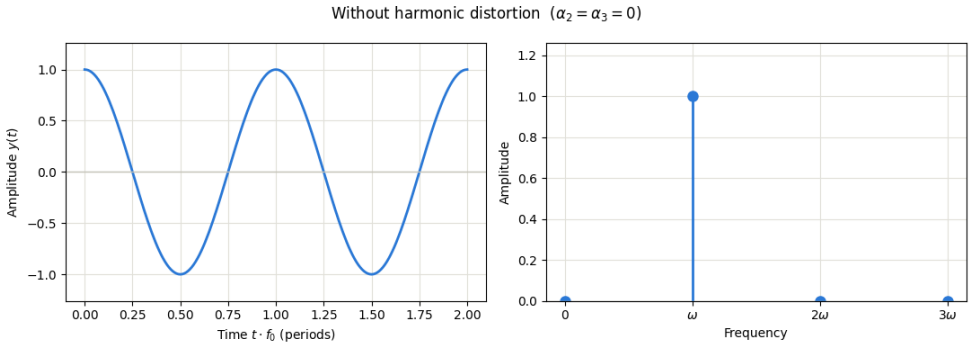
On the contrary, a memoryless nonlinear system can be approximated by the power series, [29]

$$y(t) = \alpha_1 x(t) + \alpha_2 x^2(t) + \alpha_3 x^3(t) + \cdots , \tag{2.10}$$

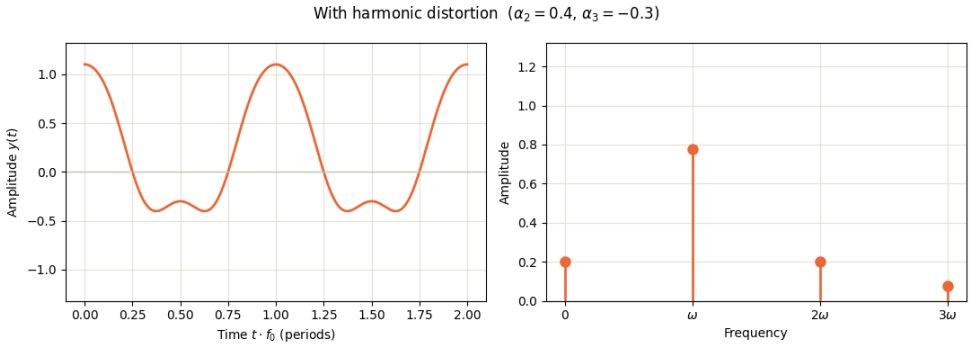
where it can be seen that the output $y(t)$ no longer shares a proportional change according to the input $x(t)$. All electronic circuits are inherently nonlinear – this is a fundamental truth of electronic engineering [18] – as semiconductor materials used to build electronic components, such as transistors and diodes, do not scale their behaviour proportionally with input applied to them. The power amplifier of a transmission chain is of course not free from this truth, and existing models that attempt to describe its nonlinear behaviour are discussed in Section 2.3. Whereas small signal operation of analogue and RF circuits can be approximated as linear, nonlinearities engender interesting effects that are not accounted for by the small-signal model. The remainder of this section elaborates on the most common phenomena observed for memoryless systems. To do so, we first describe a memoryless nonlinear system as [29]

$$y(t) \approx \alpha_1 x(t) + \alpha_2 x^2(t) + \alpha_3 x^3(t). \tag{2.11}$$

Equation (2.11) uses a third order polynomial; however as seen in (2.10), the power series can be extended for higher orders. The higher the order, the more precisely the equation can model the nonlinear behaviour of the system. The reader is cautioned that the effects of higher order nonlinearity terms and storage elements (introducing dynamic nonlinearity) are examined to determine whether (2.11) is a plausible representation [29]. On that note, depending on operating region (bias and signal level), the degree of nonlinearity of a system can be characterised as either "weakly" or "strongly" nonlinear [33]. In practice, a power amplifier must operate well below saturation to be considered weakly nonlinear, wheareas a power amplifier operating close to the 1 dB compression point (introduced in Subsection 2.2.2) is usually considered strongly nonlinear [21]. The third order model is assumed to be a sufficient approximation for weakly nonlinear systems and is chosen for ease of instruction to explain the following effects.



(a) Without harmonic distortion, i.e. $\alpha_2 = 0$ and $\alpha_3 = 0$.



(b) With harmonic distortion, where $\alpha_2 = 0.4$ and $\alpha_3 = -0.3$.

Figure 2.2: Simulation plots of the time and frequency domain, with and without harmonic distortion.

2.2.1 Harmonics

The first phenomenon is harmonic distortion. When a sinusoid is applied to a nonlinear system, the output exhibits frequency components at integer multiples of the input frequency [29] – these are called harmonics. To demonstrate this, we apply a sinusoidal input of $x(t) = A\cos(\omega t)$ to equation (2.11), then

$$y(t) = \alpha_1 A\cos\omega t + \alpha_2 A^2\cos^2\omega t + \alpha_3 A^3\cos^3\omega t \tag{2.12}$$

$$= \alpha_1 A\cos\omega t + \frac{\alpha_2 A^2}{2}(1+\cos 2\omega t) + \frac{\alpha_3 A^3}{4}(3\cos\omega t + \cos 3\omega t) \tag{2.13}$$

$$= \frac{\alpha_2 A^2}{2} + \left(\alpha_1 A + \frac{3\alpha_3 A^3}{4}\right)\cos\omega t + \frac{\alpha_2 A^2}{2}\cos 2\omega t + \frac{\alpha_3 A^3}{4}\cos 3\omega t. \tag{2.14}$$

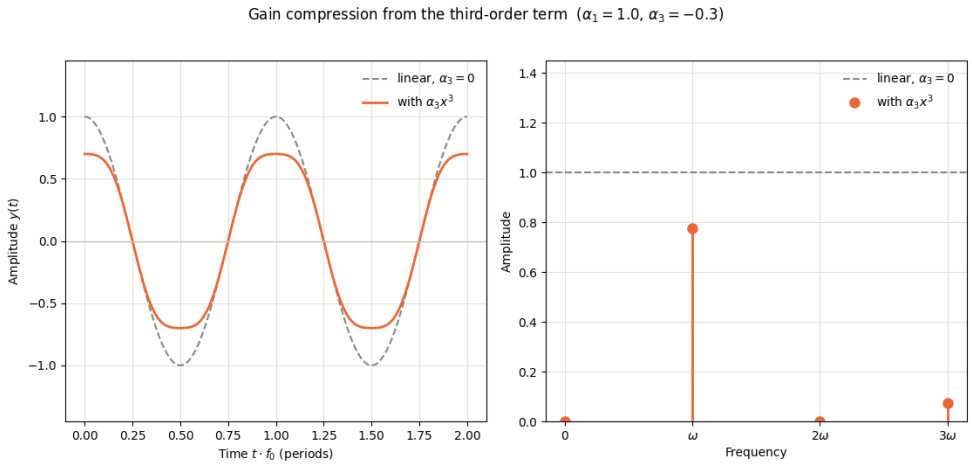


Figure 2.3: Gain compression in time and frequency domain caused by third-order nonlinearity with $\alpha_3 = -0.3$

In (2.14), the first leftmost term is a dc term. The second term is the "fundamental", while the second and third terms are the second and third harmonics respectively. From the above expansion, we can make several observations. Firstly, the second-oder nonlinearity produces a dc term, as the first term contains the second-order coefficient α_2. Secondly, each order n of nonlinearity generates a harmonic at frequency $n\omega$. Lastly, the amplitudes of the second and third harmonic are proportional to A^2 and A^3 respectively, hence the nth harmonic grows proportional to A^n.

Figure 2.2 depicts the harmonic distortion visually. In the ideal linear case shown in Figure 2.2a, there is a single peak located at the fundamental freqeuncy ω. However, the nonlinear case of Figure 2.2b illustrates energy being distributed to the dc term and harmonics at frequencies 2ω and 3ω – components that are created by nonlinearity.

2.2.2 Gain Compression

Another effect of nonlinearity is gain compression – where output gain decreases with increasing input amplitude [29]. When looking at the fundamental term in (2.14), we see that the gain experienced by $A\cos\omega t$ is equal to $\alpha_1 + \frac{3\alpha_3 A^2}{4}$. This gain is dependent on the first and third-order coefficients α_1 and α_3. Analytically, if $\alpha_1\alpha_3 > 0$, then $\alpha_1 x + \alpha_3 x$ overwhelms $\alpha_2 x$ for large x regardless of the sign of α_2, leading to an "expansive" characteristic. Inversely, for $\alpha_1\alpha_3 < 0$, a "compressive" characteristic is seen

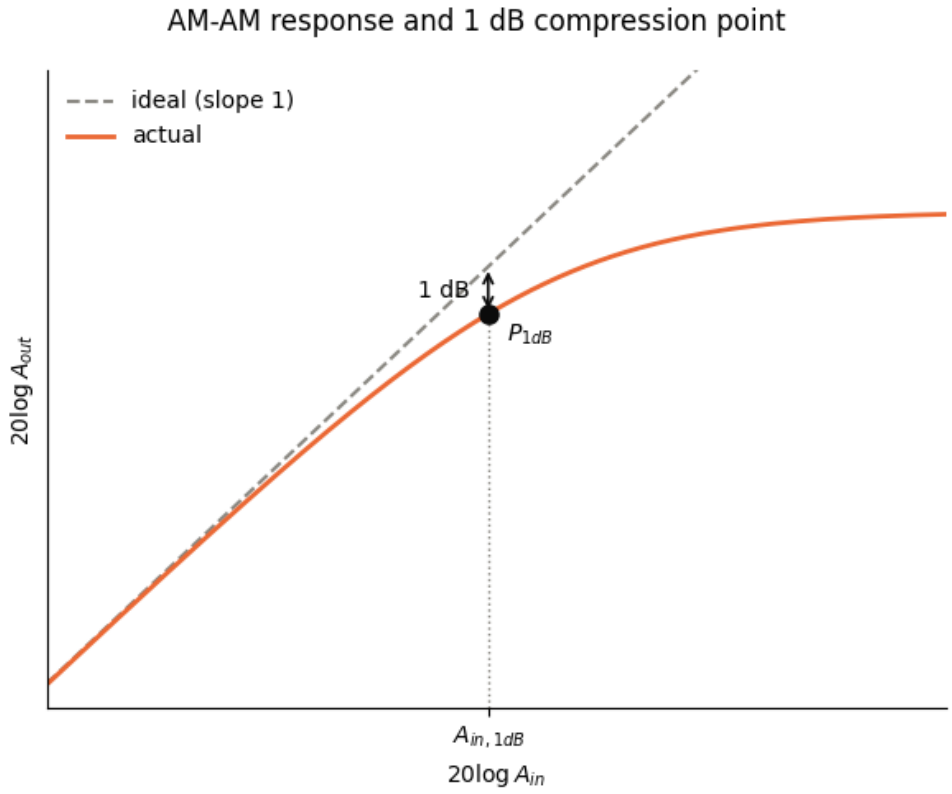


Figure 2.4: 1 dB Compression Point

for sufficiently large x. Figure 2.3 graphically demonstrates the compressive behaviour for $y(t) = \alpha_1 x(t) + \alpha_3 x^3(t)$ – simplifying to see the effect of the third-order nonlinearity term. As before, we see a harmonic produced at 3ω, but more importantly, we see the third-order term compressing the peak amplitude of the output signal in both time and frequency domain.

This compression is quantified by the 1 dB compression point [29]. The 1 dB compression point is defined as the point where the actual output gain deviates from the ideal output gain by 1 dB, typically where the output begins to near saturation. The 1 dB compression point is graphically depicted in Figure 2.4.

To obtain the 1 dB compression point mathematically, we equate the compressed gain of the fundamental term, $\alpha_1 + \frac{3\alpha_3 A_{\text{in}}^2}{4}$, to 1 dB less than the ideal gain, α_1. This results in equation (2.15).

$$20\log\left|\alpha_1 + \frac{3}{4}\alpha_3 A_{\text{in,1dB}}^2\right| = 20\log|\alpha_1| - 1\,\text{dB}. \tag{2.15}$$

From (2.15), the 1 dB compression point with reference to input amplitude is

$$P_{1\,\text{dB}} = A_{\text{in,1dB}} = \sqrt{0.145\left|\frac{\alpha_1}{\alpha_3}\right|}. \tag{2.16}$$

2.2.3 Desensitisation and Blocking

Gain compression elucidated in Subsection 2.2.2 has a further side-effect of decreasing system sensitivity - called "desensitisation" [18]. This means that weak received signals may be blocked by strong interference signals at different frequencies. This can occur when a receiver has to process a weak signal radiated from the transmitter alongside a strong signal emitted from another nearby transmitter [37]. Such decrease in sensitivity from saturation is another detrimental nonlinearity effect.

2.2.4 Intermodulation

Another highly prevalent nonlinearity effect is intermodulation. This occurs when two signals with different frequencies ω_1 and ω_2 are applied to a nonlinear system, where the output then exhibits frequency components that are not harmonics. This phenomenon arises from the "mixing" (multiplication) of the two signals [29]. The result is components generated at the new frequencies described by (2.17) [15].

$$\omega = m\omega_1 \pm n\omega_2. \tag{2.17}$$

In (2.17), ω is the new frequency, where ω_1 and ω_2 are the original frequencies. m and n are positive integers including zero, and $m+n$ is equal to the order of distortion. As such, using equation (2.11), we determine that the second-order nonlinearity generates intermodulation products at $\omega_1 \pm \omega_2$ and third order nonlinearity at $2\omega_1 \pm \omega_2$ and $2\omega_2 \pm \omega_1$.

To understand the generation of new components mathematically, we take a two-tone input signal $x(t) = A_1\cos\omega_1 t + A_2\cos\omega_2 t$ and apply it to (2.11), resulting in

$$\begin{aligned} y(t) = \alpha_1(A_1\cos\omega_1 t + A_2\cos\omega_2 t) + \alpha_2(A_1\cos\omega_1 t + A_2\cos\omega_2 t)^2 \\ + \alpha_3(A_1\cos\omega_1 t + A_2\cos\omega_2 t)^3. \end{aligned} \tag{2.18}$$

If we expand the right-hand side of equation (2.18), and discard dc terms, harmonics and second-order components at $\omega_1 \pm \omega_2$, we obtain the third-order intermodulation products in equations (2.19) and (2.20),

$$\omega = 2\omega_1 \pm \omega_2 : \quad \frac{3\alpha_3 A_1^2 A_2}{4}\cos(2\omega_1 + \omega_2)t + \frac{3\alpha_3 A_1^2 A_2}{4}\cos(2\omega_1 - \omega_2)t \tag{2.19}$$

$$\omega = 2\omega_2 \pm \omega_1 : \quad \frac{3\alpha_3 A_1 A_2^2}{4}\cos(2\omega_2 + \omega_1)t + \frac{3\alpha_3 A_1 A_2^2}{4}\cos(2\omega_2 - \omega_1)t, \tag{2.20}$$

along with the the fundamental components of equation (2.21).

$$\begin{aligned} \omega = \omega_1,\ \omega_2 : \quad & \left(\alpha_1 A_1 + \frac{3}{4}\alpha_3 A_1^3 + \frac{3}{2}\alpha_3 A_1 A_2^2\right)\cos\omega_1 t \\ & + \left(\alpha_1 A_2 + \frac{3}{4}\alpha_3 A_2^3 + \frac{3}{2}\alpha_3 A_2 A_1^2\right)\cos\omega_2 t \end{aligned} \tag{2.21}$$

Figure 2.5 graphically depicts the intermodulation products. The third-order intermodulation products at $2\omega_1 - \omega_2$ and $2\omega_2 - \omega_1$ are of particular interest, as they appear directly in the vicinity of ω_1 and ω_2. These corrupt the transmitted signal - often lying within the transmission bandwidth. A common measure of the extent of third-order intermodulation products is the "third intercept point" (IIP_3) [29]. This is defined as the point where the amplitude of the intermodulation products equals that of the fundamental tones at the output. This is demonstrated graphically in Figure 2.6. In practice, this point is usually never reached, as the output typically reaches saturation before this.

To derive the third intercept point mathematically, the amplitude of the third-order intermodulation products in (2.19) and (2.20) can first be simplified for input tones of the same amplitude, i. e. $A = A_1 = A_2$, to $\frac{3}{4}\alpha_3 A^3$. At the input at which this theoretical

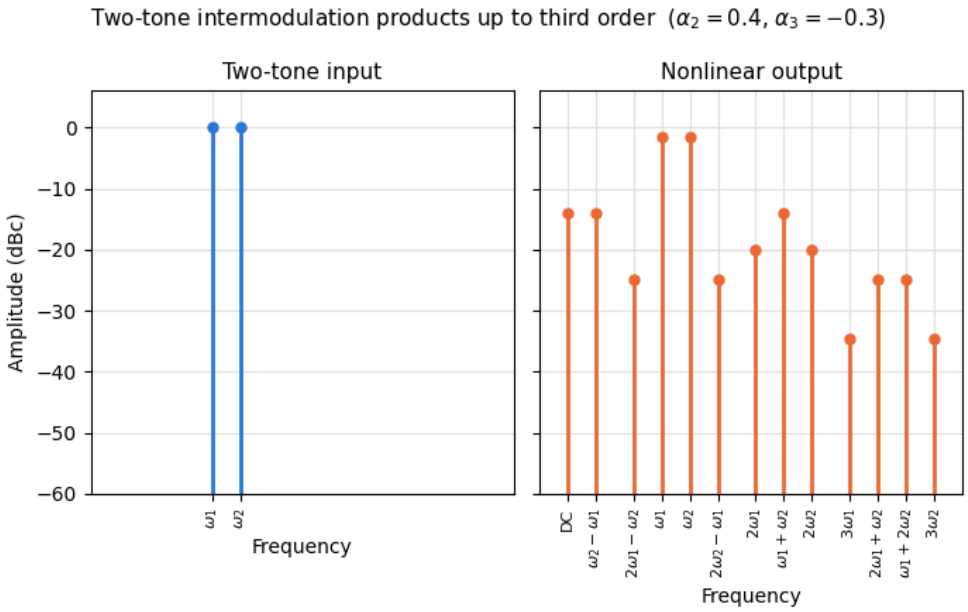


Figure 2.5: Two-tone intermodulation products up to third-order with $\alpha_2 = 0,4$ and $\alpha_3 = -0,3$

input third intercept point ($IIP3$) occurs, the uncompressed linear component $\alpha_1 A$ would equal the third order intermodulation product, resulting in

$$|\alpha_1 A_{\text{IIP3}}| = \left|\frac{3}{4}\alpha_3 A_{\text{IIP3}}^3\right|, \tag{2.22}$$

solving for $A_I IP3$ yields the input third intercept point ($IIP3$)

$$A_{\text{IIP3}} = \sqrt{\frac{4}{3}\left|\frac{\alpha_1}{\alpha_3}\right|}. \tag{2.23}$$

2.2.5 Phase Distortion

The final effect of nonlinearity we discuss is phase distortion. Sections 2.2 has thus far only covered effects that involves nonlinear changes to the output amplitude with respect to input amplitude. This is commonly referred to as AM-AM conversion. However, most power amplifier circuits exhibit memory effects due to storage components

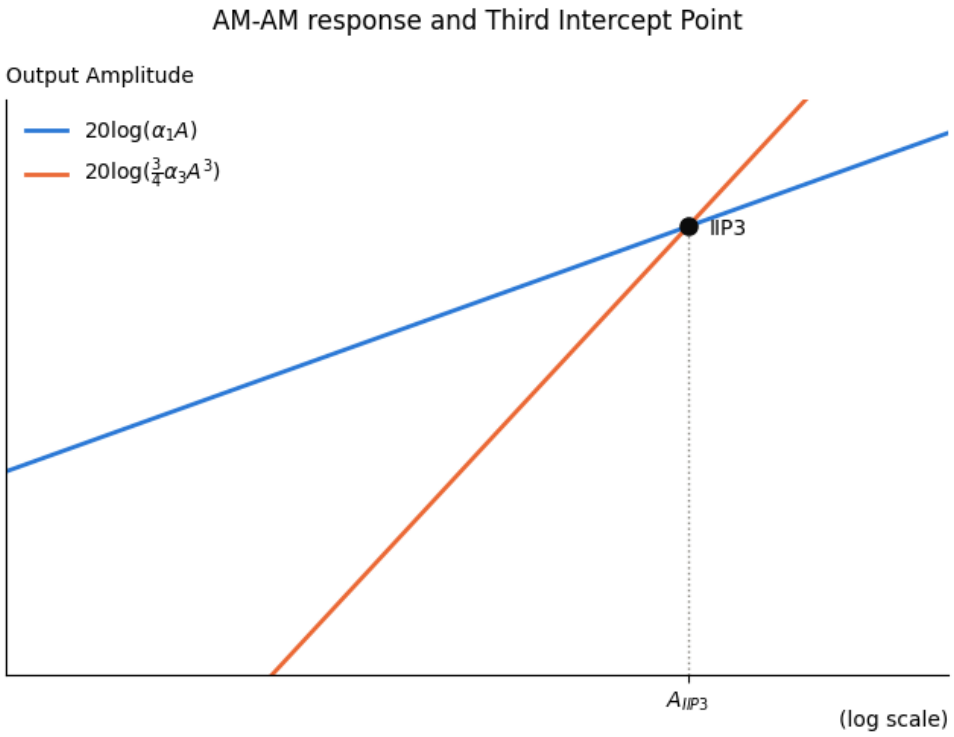


Figure 2.6: Third Order Intercept Point

such as capacitors and inductors, resulting in phase distortion from capacitive and inductive nonlinearities. This effect is commonly referred to as AM-PM conversion [21]. The cause behind AM-PM conversion can be understood by taking the fundamental component from (2.14) seen in (2.24) [18].

$$\left(\alpha_1 A + \frac{3\alpha_3 A^3}{4}\right)\cos\omega t \tag{2.24}$$

The input amplitude is denoted by A, while α_1 and α_3 are the first and third order coefficients respectively. The amplitude of equation (2.24) consists of the first order component $\alpha_1 A$ and the third order component $\frac{3\alpha_3 A^3}{4}$. Due to nonlinear capacitance, a phase difference can exist between these two components. The response is then a vector sum of two phasors illustrated in (2.25).

$$V = \alpha_1 A + \frac{3}{4}\alpha_3 A^3 e^{j\theta}. \tag{2.25}$$

In (2.25), α_1 and α_3 are again the first and third order coefficients respectively. The sum of the phasors is denoted V and the phase difference between the first (linear) and third order term is denoted by θ. As such, $\frac{3}{4}\alpha_3 A^3$ possesses a different phase from the linear component. Addition of the two phasors, linear and third order, leads to a phasor V with a new angle. This new angle is dependent on variations in the input amplitude A, regardless of whether θ remains constant [18]. Hence, changes in input amplitude result in changes in output phase of the power amplifier, namely AM-AM conversion. This becomes more critical as the power amplifier is driven to higher input amplitudes into saturation, because the third order component increases commensurately with A^3.

2.3 Power Amplifier Nonlinearity Models

Section 2.2 described the effects that nonlinearity produces, but not how the nonlinear behaviour of a particular power amplifier is captured in a form suitable for simulation. Two broad approaches exist. Physical models derive the response from the geometry, material parameters and equivalent circuit of the transistor. Such models are accurate, but they are computationally expensive and demand device knowledge that manufacturers rarely disclose. Behavioural models instead treat the power amplifier as a black box and describe only the relationship between the input and output signal envelopes [32]. Because this thesis is concerned with system level effects rather than device physics, and because the parameters of a behavioural model can be extracted from the limited data published in manufacturer datasheets, behavioural models are the appropriate choice and form the subject of this section.

A memoryless behavioural model is fully characterised by two functions of the input envelope amplitude. The first, $A(r)$, returns the output amplitude for a normalised input amplitude r and therefore describes AM-AM conversion. The second, $\Phi(r)$, returns the output phase shift and describes the AM-PM conversion introduced in Subsection 2.2.5. Each model below is accordingly presented as a pair of such expressions, whose coefficients are obtained by curve fitting against measured or datasheet characteristics. It should be noted that all models presented here are memoryless, and consequently repro-

duce neither frequency dependent behaviour nor the long term memory effects caused by thermal and bias network dynamics.

2.3.1 Saleh Model

The earliest and most widely cited behavioural model was proposed by Saleh for travelling wave tube amplifier (TWTA) devices [31]. Both conversion characteristics are expressed as rational functions of the input amplitude,

$$A(r) = \frac{\alpha_a r}{1 + \beta_a r^2}, \tag{2.26}$$

$$\Phi(r) = \frac{\alpha_\phi r^2}{1 + \beta_\phi r^2}, \tag{2.27}$$

where α_a and β_a determine the small signal gain and the saturation level of the amplitude characteristic, while α_ϕ and β_ϕ determine the corresponding phase characteristic. The model requires only four parameters, which is a large part of its appeal. Its shortcoming is that it was derived for travelling wave tubes, whose compression characteristic differs markedly from that of a solid state device. Consequently the Saleh model represents solid-state power amplifier (SSPA) behaviour poorly [8], which motivated the models that followed.

2.3.2 Ghorbani and Sheikhan Model

Ghorbani and Sheikhan proposed a model specifically for SSPAs [9]. It retains the rational structure of the Saleh model but raises the input amplitude to a fitted exponent and adds a linear term, giving

$$A(r) = \frac{x_1 r^{x_2}}{1 + x_3 r^{x_2}} + x_4 r, \tag{2.28}$$

$$\Phi(r) = \frac{y_1 r^{y_2}}{1 + y_3 r^{y_2}} + y_4 r. \tag{2.29}$$

The additional exponents x_2 and y_2 allow the sharpness of the transition into compression to be adjusted independently of the saturation level, and the linear terms $x_4 r$ and $y_4 r$ correct the slope in the small signal region. The eight parameters give the model appreciably more freedom than the Saleh model, and it fits solid state devices correspondingly better [8]. The price is that the characteristic is known to "roll over" beyond the fitted range, that is, the predicted output amplitude begins to decrease once the input is driven past the region covered by the fitting data. This makes the model unreliable when extrapolated into deep saturation.

2.3.3 Rapp Model

The model proposed by Rapp is the one most commonly encountered in the SSPA literature [28]. Its AM-AM characteristic is

$$A(r) = \frac{g\, r}{\left[1 + \left(\frac{g\, r}{A_{\text{sat}}}\right)^{2p}\right]^{\frac{1}{2p}}}, \tag{2.30}$$

where g is the small signal gain, A_{sat} the saturated output amplitude and p the smoothness factor, which controls how abruptly the characteristic transitions from the linear region into saturation. The structure of (2.30) is what distinguishes the model: the output tends towards $g\, r$ for small r and asymptotically towards A_{sat} for large r, so the characteristic is monotonic and saturates gracefully by construction rather than as a consequence of the fit. As $p \to \infty$ the model approaches an ideal limiter.

Rapp provided no AM-PM expression, arguing that the phase conversion of a solid state amplifier is small enough to be neglected [28]. That assumption does not hold for the devices of interest here, and Honkanen and Häggman subsequently supplied a phase characteristic intended to be used alongside (2.30) [11],

$$\Phi_{\text{HH}}(r) = \frac{b}{1 + e^{-c(r-a)}} \cdot \frac{1}{p_1 r + p_2}, \tag{2.31}$$

in which a is the cut-off input voltage at which the transistor begins to conduct, b the switch-on step magnitude in degrees, c the steepness of that step, and p_1 and p_2

the parameters adjusting the descending slope. The logistic first factor produces the switch-on step and the second produces the subsequent descent.

Equation (2.31) is strictly negative-going and bounded above by zero degrees, so it cannot represent a device whose phase shift is positive at low drive levels. Fisher and Al-Sarawi addressed this by adding an offset parameter d [8],

$$\Phi_{\mathrm{F}}(r) = \frac{b}{1 + e^{-c(r-a)}} \cdot \frac{1}{p_1 r + p_2} - d, \tag{2.32}$$

which translates the whole characteristic vertically and thereby admits positive phase shifts. Although the coefficients of (2.31) were originally related to bipolar transistor parameters, the model is reported to perform well for laterally diffused metal oxide semiconductor (LDMOS) devices also [8].

2.3.4 Later Refinements

Two further refinements are relevant. White, Burr and Javornik proposed a model built from exponential terms rather than a rational function [38],

$$A_W(r) = a\left(1 - e^{-br}\right) + c\, r\, e^{-dr^2}, \tag{2.33}$$

$$\Phi_W(r) = \begin{cases} f\left(1 - e^{-g(r-h)}\right), & r \geq h, \\ 0, & r < h, \end{cases} \tag{2.34}$$

where a and b set the saturation level and the small signal gain while c and d adjust the shape of the characteristic within the operating region. The phase characteristic is defined piecewise, remaining zero until the threshold h is exceeded.

Fisher and Al-Sarawi were unable to reproduce the published results using (2.33) and concluded that it contains a typographical error. Placing the third term inside the bracket scaled by a, so that

$$A_{W,\mathrm{corr}}(r) = a\left[1 - e^{-br} + c\, r\, e^{-dr^2}\right], \tag{2.35}$$

recovers the accuracy originally reported [8]. The distinction is not merely cosmetic: with (2.33) the characteristic rolls over beyond the fitted range in the manner of the Ghorbani and Sheikhan model, whereas (2.35) behaves as expected of a solid state device.

O'Droma, Meza and Lei proposed the modified Saleh model, developed for LDMOS based amplifiers [26], given by

$$A_{MS}(r) = \frac{\alpha_a r^{\eta}}{(1+\beta_a r^{\gamma})^{\nu}} - \varepsilon, \tag{2.36}$$

$$\Phi_{MS}(r) = \frac{\alpha_\phi r^{\eta}}{(1+\beta_\phi r^{\gamma})^{\nu}} - \varepsilon. \tag{2.37}$$

Comparison with (2.26) and (2.27) shows the nature of the modification: the exponents η, γ and ν generalise the fixed powers of the original Saleh model, and the offset ε shifts the characteristic. The additional degrees of freedom let the compact rational form of the original follow a solid state device far more closely, and the model achieves the best fit of all those considered when measurement data extending into saturation is available [8].

2.3.5 Model Comparison and Selection

Fisher and Al-Sarawi compared the above models by fitting each to the same measured device data and evaluating the normalised mean squared error (NMSE) of the resulting AM-AM characteristic [8]. Their results are reproduced in Table 2.1, where a more negative figure indicates a closer fit.

Two conclusions follow. Firstly, the Saleh model is clearly the weakest for solid state devices, which is consistent with its origin as a travelling wave tube model. Secondly, the ranking depends on how far the fitting data extends. When only data below saturation is available, which is the usual situation when parameters are extracted from a datasheet, the Rapp model of (2.30) extrapolates into the saturation region more reliably than the models that fit the measured range more tightly [8]. When measurement data extending into saturation is available, the modified Saleh model is the most accurate in that region.

Table 2.1: AM-AM modelling accuracy of behavioural power amplifier models, after Fisher and Al-Sarawi [8].

Model	NMSE (dB)
Saleh [31]	−27.13
Ghorbani and Sheikhan [9]	−35.44
Rapp [28]	−42.33
White et al. [38]	−36.95
White et al., corrected [8]	−44.00
Modified Saleh, O'Droma et al. [26]	−48.54

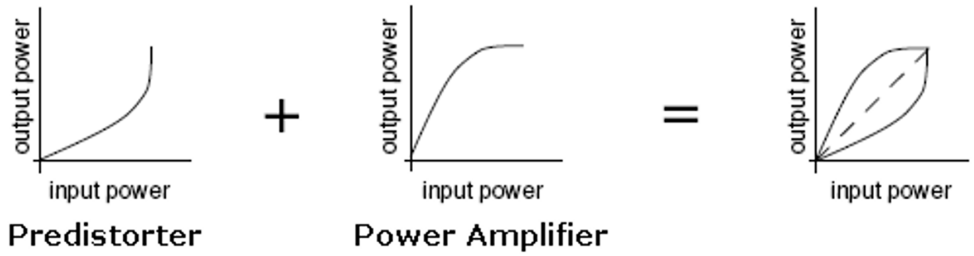


Figure 2.7: The basic idea of predistortion [20]

2.4 Predistortion

This section introduces the concept of predistortion and the most common predistortion techniques used to mitigate nonlinearity effects. The idea of predistortion is to insert a nonlinear element (predistorter) proceeding a power amplifier, so that the cascaded transfer characteristics of the two is linear. The predistorter approximates the inverse characteristic of the power amplifier, thereby essentially cancelling out the nonlinearity at the output by first overcorrecting the input – this underlying idea is illustrated in Figure 2.7. The result is a linear input-output relationship of the predistorter-amplifier combination. According to the position of the predistorter in the transmitter, predistortion is divided into three categories [20]:

- RF Predistortion
- IF Predistortion
- Baseband Predistortion

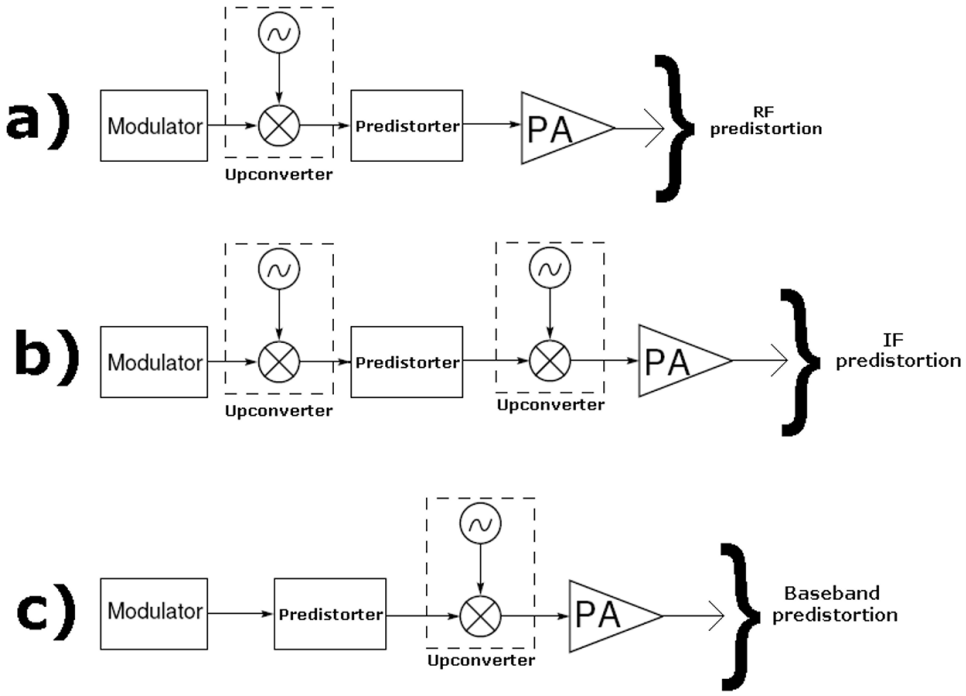


Figure 2.8: Types of predistortion: RF predistortion a), IF predistortion b), baseband predistortion c) [20]

The position of the predistorter block for each of the three categories is depicted in Figure 2.8. In RF predistortion, the predistorter element must operate at the final carrier frequency [13]. Whereas for IF predistortion, the predistorter operates at the more convenient IF frequency – allowing the predistorter block to be reused at different carrier frequencies [13]. Finally, in baseband predistortion, the predistorter block operates at the baseband frequency, and the predistorted signal is upconverted up until the final carrier frequency.

Power amplifier characteristics can be affected and changed by temperature changes, aging, or even output load changes [5]. To preserve linearity for such cases, a predistorter should also adjust for this correspondingly. This divides predistortion into two further categories: “adaptive” and “non-adaptive” predistortion [20]. In adaptive predistortion, information on the power amplifier output is used in combination with its input signal to continually adjust the predistorter characteristic [20]. Adaptive predistortion can be applied to all RF predistortion, IF predistortion, and baseband predistortion categories, though is most commonly seen at baseband. Such is called “adaptive baseband predistortion”, which is a highly popular linearisation method today [5].

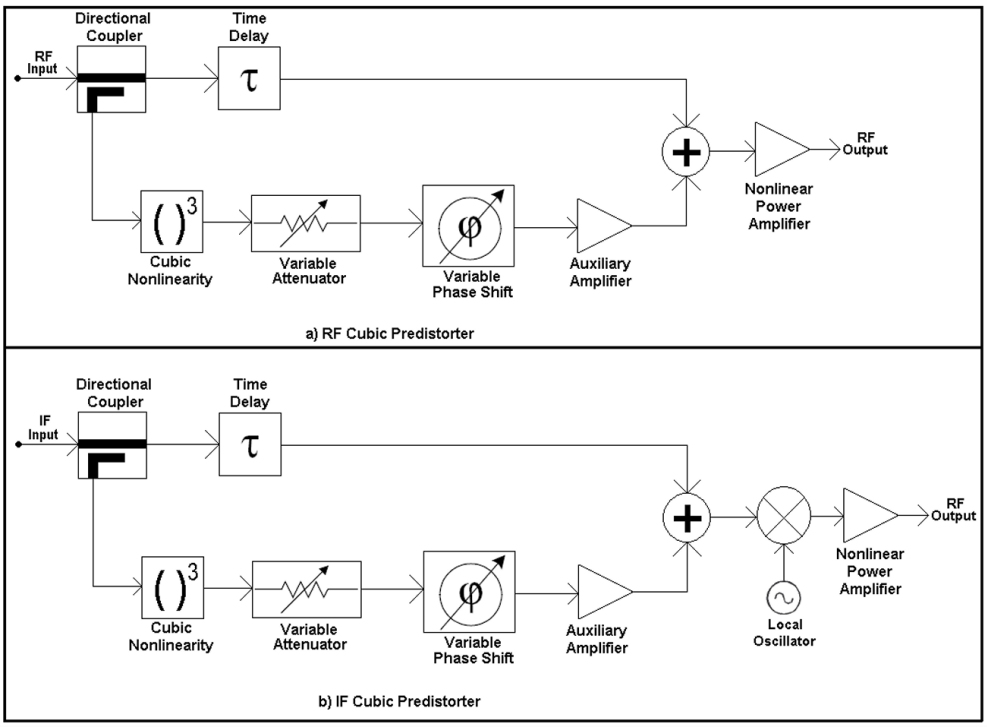


Figure 2.9: a) RF cubic predistorter b) IF cubic predistorter [5]

2.4.1 RF and IF Predistortion

RF and IF predistortion are similar in operation, differing only in the frequency at which the predistorter acts. As RF predistortion operates at the final carrier frequency, it is difficult to make adaptive, and such predistorters are consequently non-adaptive in general – requiring the nonlinearity to be cancelled to be known in advance [5]. Its principal advantage is its simple implementation. Operating at IF instead lowers the frequency of operation, which permits the use of components that cannot work with RF signals [5].

The fundamental advantage of both is their ability to linearise the entire bandwidth of the amplifier, making them well suited to wideband multicarrier systems [5, 13]. The degree of linearity improvement attainable depends primarily on the transfer characteristic of the power amplifier itself – the better behaved this characteristic is, the greater the improvement that can be achieved and maintained across varying input conditions [13].

The aim is therefore to realise a circuit whose transfer function approximates the inverse of the power amplifier characteristic, such that the combination of the two produces a linear input-output relationship. A widely used configuration for this purpose is the cubic predistorter depicted in Figure 2.9, which is based on the cancellation of third-order intermodulation products [5]. The input signal is split into two paths. The upper path merely delays the signal to synchronise it with the lower path. The lower path contains a cubic nonlinearity, a variable phase shifter and attenuator, and an auxiliary amplifier. The cubic element provides a compressive characteristic, the phase shifter and attenuator adjust phase and amplitude such that the distortion cancels, and the auxiliary amplifier compensates for the considerable attenuation introduced by the cubic element – without contributing distortion of its own, as its input power level is low [5].

Assuming both paths are synchronised, the upper and lower path signals are given by

$$v_{\text{upper}} = a_1 v_{\text{in}}, \tag{2.38}$$

$$v_{\text{lower}} = a_2 v_{\text{in}} - b v_{\text{in}}^3, \tag{2.39}$$

where (2.39) exhibits the compressive characteristic of the cubic element. Subtracting the lower path from the upper path yields the predistorter output

$$v_{\text{pd}} = (a_1 - a_2)\, v_{\text{in}} + b v_{\text{in}}^3, \tag{2.40}$$

which is an expansive characteristic with a linear gain of $a_1 - a_2$. By appropriate choice of a_1, a_2 and b, (2.40) can be used to predistort any compressive power amplifier characteristic [5, 13]. The cubic nonlinearity itself is commonly realised using a single diode, a pair of anti-parallel diodes, or field-effect transistors [5].

2.4.2 Baseband Predistortion

In baseband predistortion, the predistorter element operates at baseband frequencies. This serves as an advantage, as it is significantly easier to implement the inverse characteristic of the nonlinearity at baseband due to implementation simplicity and reconfigurability. This can commonly be implemented in firmware on digital signal processing

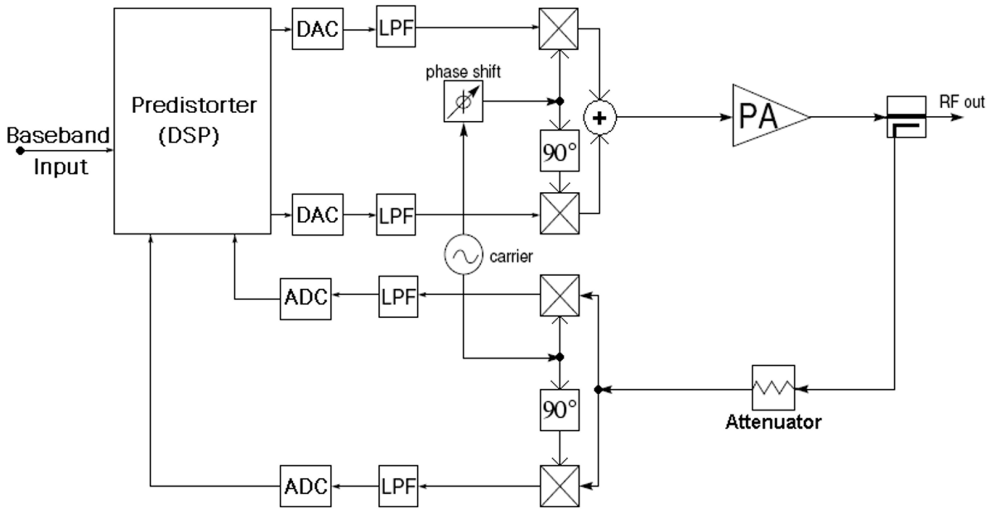


Figure 2.10: Adaptive digital baseband predistortion [20]

devices such as digital signal processors (DSPs), field-programmable gate arrays (FPGAs), and application-specific integrated circuits (ASICs) [20] – which can be flexibly reconfigured. Thus, when the term baseband predistortion is used, it generally refers to "digital baseband predistortion". A further advantage of digital baseband predistortion is that it is rather straightforward to implement adaptive predistortion – an example of this is seen in Figure 2.10. As depicted, the baseband input signal is fed into the predistorter, which outputs the predistorted digital signal. This signal is then converted by a DAC to an analogue signal, then upconverted to RF, and finally amplified by the power amplifier for transmission. The adaptive nature is shown as the power amplifier output is sampled, downconverted to baseband, converted to digital form by an ADC, and continuously used to adapt the predistorter characteristic [20].

Digital baseband predistortion can be divided into the following groups. Among several techniques, digital baseband predistortion can be split into three implementation approaches – namely look-up table (LUT), polynomial model, and neural network based predistorters [7]. Whereas neural network based predistorters can excellently approximate the nonlinear inverse characteristic, their training process is rather arduous and time consuming. Furthermore, hardware implementation is difficult in practical applications [7, 25, 22]. LUT predistorters are simple to implement, but their linearisation performance depends on the size of the LUT [7, 16, 35, 34, 17]. Polynomial model based predistortion is also widely common – including well known models such as the Volterra

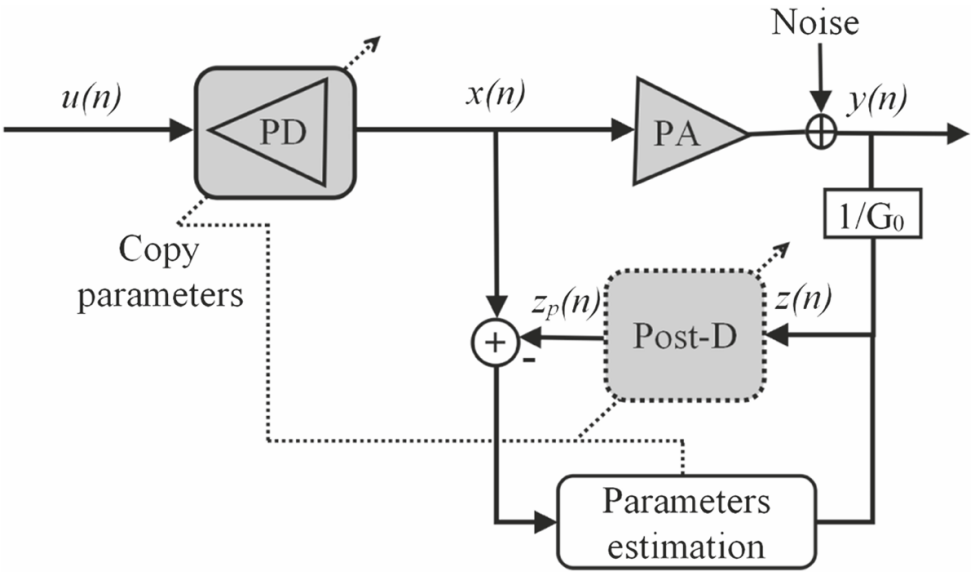


Figure 2.11: Indirect learning architecture [7]

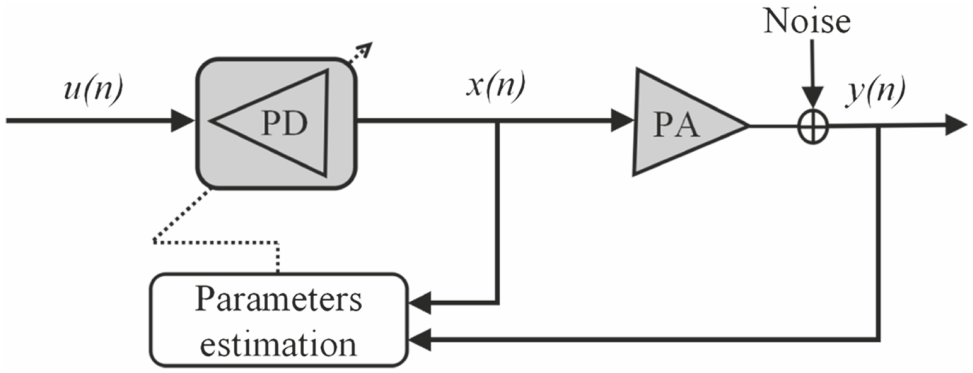


Figure 2.12: Direct learning architecture [7]

series [6], model polynomial [4, 14, 16], generalised model polynomials [24], and etc – with tradeoffs between different models in accuracy and computational complexity.

Digital baseband predistortion can be realised by three main learning approaches. Two of these approaches involve computing models of the power amplifier and its postinverse [1, 23] – namely indirect learning architecture (ILA) and direct learning architecture (DLA). The last approach, namely iterative learning control (ILC) [2], aims to iteratively learn the required predistorted power amplifier input itself – forgoing modelling of the power amplifier and its parameters. All these approaches can be implemented

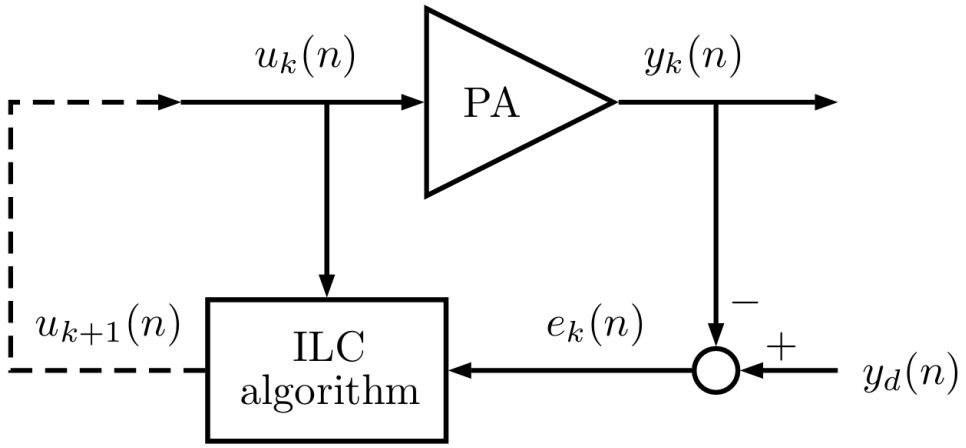


Figure 2.13: Iterative learning control block diagram [10]

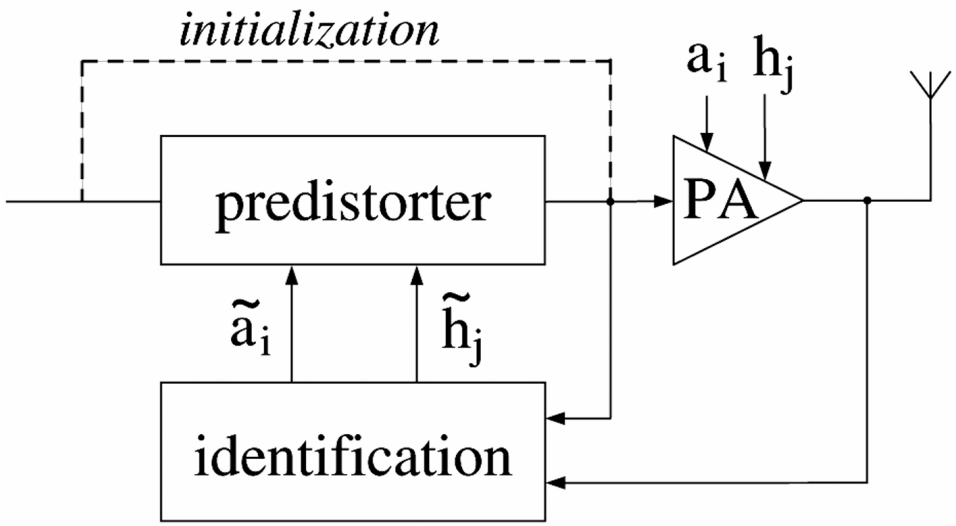


Figure 2.14: Model of adaptive predistortion

using the LUT, polynomial model, and neural network based approaches [5]. Both architectures have different tradeoffs. Whereas the ILA is simpler by directly calculating the predistorter linearising function, the DLA approach is more robust against noise at the power amplifier chain [1]. The following sections 2.4.2.1 and 2.4.2.2 elaborate on the two architectures and how they each function.

2.4.2.1 Indirect Learning Architecture

In ILA [4], the inverse of the power amplifier nonlinearity is first determined using the input and scaled down output of the power amplifier. This model is then forwarded to the predistorter block. This is seen in Figure 2.11, where the postdistorter (Post-D) block calculates the nonlinearity inverse. The postdistorter is iteratively estimated by an identification algorithm such as the least squares method, aiming to minimise the error between power amplifier input and scaled output with each iteration. Once identified, the postdistorter block is essentially copied to the front of the power amplifier as the predistorter.

The postdistorter is commonly modelled as a memory polynomial [4], whose output is given by

$$z_p(n) = \sum_{k=0}^{K} \sum_{l=0}^{L} a_{kl}\, \Phi_{kl}[z(n)]\,, \tag{2.41}$$

where $z(n) = \frac{y(n)}{G_0}$ is the scaled-down power amplifier output, $\Phi_{kl}[z(n)] = z(n-k)|z(n-k)|^{2l}$, G_0 the desired gain, a_{kl} the complex-valued coefficients, K the memory depth and $2L+1$ the highest nonlinearity order [7]. In a perfect system, $z_p(n) = x(n)$ is required.

Assuming a total of N samples, (2.41) is expressed in matrix form as

$$\mathbf{z}_p = \mathbf{Z}\mathbf{a}, \tag{2.42}$$

where $\mathbf{z}_p$ collects the postdistorter output samples, $\mathbf{Z}$ the corresponding basis functions Φ_{kl} and $\mathbf{a}$ the coefficients. The latter are then extracted by the least squares method as

$$\hat{\mathbf{a}} = \left(\mathbf{Z}^H\mathbf{Z}\right)^{-1}\mathbf{Z}^H\mathbf{z}_p, \tag{2.43}$$

where $(\cdot)^H$ denotes the complex conjugate transpose operation [7]. Once the postdistorter model is identified, it is copied and placed in front of the power amplifier as the predistorter.

2.4.2.2 Direct Learning Architecture

In DLA [14, 39, 3], the behaviour of the power amplifier is first modelled using its input and output. It is again estimated iteratively using the least squares method for instance. Once identified, the power amplifier model is reversed to determine the predisorter. This is depicted in Figure 2.12.

The power amplifier is modelled as a memory polynomial

$$y(n) = \sum_{p=0}^{P} \sum_{q=0}^{Q} c_{pq}\, \Phi_{pq}[x(n)]\,, \tag{2.44}$$

where $\Phi_{pq}[x(n)] = x(n-p)|x(n-p)|^{2q}$, P is the memory depth, $2Q+1$ the highest nonlinearity order and c_{pq} the model coefficients, again extracted by the least squares method [7]. For an ideal cascaded predistorter and power amplifier system,

$$y(n) = \sum_{p=0}^{P} \sum_{q=0}^{Q} c_{pq}\, \Phi_{pq}[x(n)] = G_0 u(n), \tag{2.45}$$

where $u(n)$ denotes the input of the cascaded system and $x(n)$ the predistorted signal. The output is separated into a static part $s(n)$, depending only on the current input sample, and a dynamic part $d(n)$, depending on the previous input samples,

$$\begin{cases} y(n) = s(n) + d(n) \\ s(n) = \displaystyle\sum_{q=0}^{Q} c_{0q}\, \Phi_{0q}[x(n)] \\ d(n) = \displaystyle\sum_{p=1}^{P} \sum_{q=0}^{Q} c_{pq}\, \Phi_{pq}[x(n)] \end{cases} \tag{2.46}$$

From (2.45) and (2.46), the static part is rewritten as

$$s(n) = e^{j\angle x(n)} \sum_{q=0}^{Q} c_{0q}\, |x(n)|^{2q+1} = G_0 u(n) - d(n), \tag{2.47}$$

where $\angle x(n)$ and $|x(n)|$ are the phase and amplitude of the predistorted signal respectively. As $d(n)$ depends only on previous samples, the right-hand side of (2.47) and the coefficients c_{0q} are known at instant n. Taking the absolute value yields

$$\left| \sum_{q=0}^{Q} c_{0q} \left| x(n) \right|^{2q+1} \right| = \left| G_0 u(n) - d(n) \right| . \tag{2.48}$$

Equation (2.48) is a high-order nonlinear equation, whose solution $|x(n)|$ is found by a root-finding procedure [7]. The corresponding phase is then calculated as

$$\angle x(n) = \arg \left\{ \frac{G_0 u(n) - d(n)}{\sum_{q=0}^{Q} c_{0q} \left| x(n) \right|^{2q+1}} \right\}, \tag{2.49}$$

and the predistorted signal is finally given by

$$x(n) = |x(n)| \, e^{j \angle x(n)}. \tag{2.50}$$

Both DLA and ILA approaches involve a learning phase to model characteristics – the power amplifier for DLA or postdistorter for ILA. Both architectures can be made adaptive by periodically updating the power amplifier and postdistorter models respectively. This recomputes the weighting coefficients for the given memory depth and linearity order. This adaptive approach is illustrated in Figure 2.14, where the a and h coefficients represent nonlinearity and memory depth respectively.

Whereas the ILA approach is generally easier to implement than DLA due to its direct derivation of the nonlinear inverse characteristic, DLA is more robust in the presence of noise [1]. However, this does not matter in the context of pulse compression radar. This is because the same set of waveforms is transmitted. As every pulse is identical, the power amplifier output can be captured for many pulses and averaged; hence, the random noise averages out.

On that note, the DLA and ILA algorithms are further simplified for pulse compression radar, as the predistorted signal $x(n)$ only needs to be computed once for a recurring pulse. It can be stored and loaded for transmission, saving the computational time and workload of having to recalculate $x(n)$ for the case of varying inputs.

2.4.2.3 Iterative Learning Control

ILC is a control theory technique to iteratively learn the power amplifier input that minimises the difference between its output and the desired reference signal [2]. This is a non-model based learning approach which aims to directly find the optimal input signal to the power amplifier, serving as the predistorted signal. The derived optimal predistorted signal can also then be used a reference to extract power amplifier and predistortion modelling parameters, though the approach itself does not model the power amplifier nor predistortion. Rather, it aims to iteratively minimise the error between the power amplifier output and its theoretical ideal linear output. As such, the ILC algorithm requires that the desired output response is the same for each iteration, which is suitable for pulse compression radar [10]. The learned optimal input signal is stored and used directly as the predistorted signal, forgoing the need for a predistorter structure.

A block diagram of the ILC algorithm is depicted in Figure 2.13. $u_k(n)$ and $y_k(n)$ are the input and output signal of the power amplifier respectively for iteration k. $y_d(n)$ is the desired ideal unamplified linear output of the power amplifier. The algorithm starts by initialising both $u_1(n)$ (the first iteration) and $y_d(n)$ to the reference input signal $x(n)$ i. e. $u_1(n) = x(n)$ and $y_d(n) = x(n)$. Then, for iterations $k = 1$ to K, the following steps are performed [2]: Firstly, $u_k(n)$ is fed into the power amplifier. The output $y_k(n)$ is correspondingly captured. Then, the difference between the scaled-down captured output and desired reference output is calculated as the error $e_k(n) = y_d(n) - \frac{y_k(n)}{G}$ where G is the power amplifier gain. The error signal $e_k(n)$ is then used to update the input of the next iteration $u_{k+1}(n) = u_k(n) + e_k(n)$. With each iteration, the error decreases until an optimal predistorted power amplifier input signal is obtained.

The ILC approach is ideal for pulse compression radar, as the same set of pulse signals is transmitted. The optimal predistorted waveform for these pulses simply needs to be learned and stored by applying ILC, to then be easily loaded for transmission. The algorithm is made adaptive by periodic recomputation of the optimal waveform in the background.

3 Requirements Analysis

This chapter establishes the objectives and requirements of the present work. To provide a clear specification for the developed solution, the requirements are derived systematically through a step-by-step analysis based on the following.

3.1 Research Aim and Questions

The overall aim of this thesis is two-fold. It is to first understand the influence of power amplifier nonlinearity on the power amplifier chain of the SSR-X pulse compression radar, and secondly, to counteract these effects using digital predistortion. These nonlinearities generally compound as input power increases towards saturation. Digital predistortion can counteract these effects, enabling the power amplifier to run closer to saturation without violating regulations and a significant performance degradation. This keeps the radar spectrally compliant and enables the radar to accommodate higher power and consequently higher SNR. This increased SNR headroom enhances detection probability of the radar. For instance, this headroom can be used to increase radar detection range or detect targets with a smaller cross section at the same range – such is a direct performance gain. SNR at the output of the system is increased without the need for purchasing additional expensive hardware. Mitigating nonlinearity effects may also aid in shrinking range sidelobes around strong targets, recovering the ability to detect weaker targets close in their vicinity. Moreover, if the sidelobe level reaches a certain threshold, false targets may emerge – misleading navigators. Mitigating such effects is paramount to improving radar performance and reliability. Specific research questions pertinent to the overall objective are outlined below:

- **RQ1:** What are the effects of power amplifier chain non-linearity on the SSR-X pulse compression radar performance?

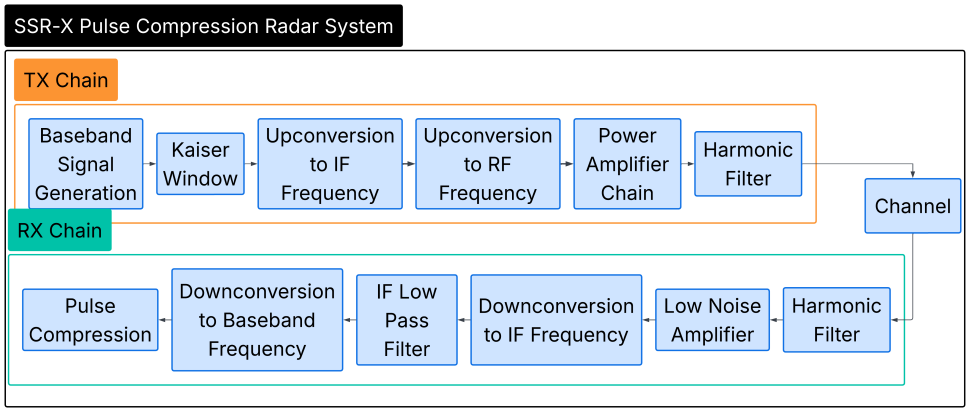


Figure 3.1: SSR-X Pulse Compression Radar System

- **RQ2:** What existing models best emulate the power amplifier chain non-linearity effects on the SSR-X pulse compression radar?
- **RQ3:** What digital predistortion techniques exist to counteract the non-linear effects of the power amplifier chain?
- **RQ4:** To what extent can digital predistortion counteract the non-linear effects of the power amplifier chain and improve performance?

3.2 System Environment

This section describes the hardware environment of the SSR-X pulse compression radar system. The system is confined to the pulse compression radar chain of the SSR-X radar transceiver, shown in Figure 3.1. A radar pulse signal is generated in the transmit chain and then input into the power amplifier for signal amplification before transmission. The echo of this pulse is then captured by the receive chain after some delay and attenuation. Pulse compression is performed at the end of the signal chain to increase SNR, increase range resolution, and determine range of a detected target. The thesis work focuses on investigating the nonlinear effects the power amplifier chain has on the system and finding a suitable digital predistortion solution to correct for these effects.

3.3 Stakeholders

The stakeholders of this thesis are those who influence the direction of work and benefit from its results. They are outlined below along with their interests:

- **NACOS Marine Radar Developers:** The engineers at NACOS Marine, who are responsible for the maintenance and development of the radar system, benefit from the results of this work, as it will provide insight on what the effects of power amplifier nonlinearity have on the radar system and its performance. It will help them understand the extent of these effects and how they may be detrimental to the overall radar system performance. The work may also provide potential solutions to these issues – guiding engineers in improving the quality and performance of the product they are developing.
- **NACOS Marine Product Management:** The product management department at NACOS Marine may benefit from this work, as it may improve the performance of the product they are selling. Higher performance – increased target range or detection of smaller targets – may open new market potential, increasing sales and revenue of the product.
- **International Electrotechnical Commission (IEC) Technical Committee:** According to the International Standard IEC 62388, X-band maritime radar shall transmit within the confines of the allocated band 9.2 GHz to 9.5 GHz [12, Sec. 6.2.1.1]. Beyond the operating band itself, the standard constrains the shape of the transmitted spectrum by means of a normative out-of-band emission mask [12, Annex B]. The necessary bandwidth of the transmission – calculated from the measured pulse width, rise time and fall time – shall be contained completely within the allocated band, which Annex B references as 9.3 GHz to 9.5 GHz for X-band radar [12, Sec. B.2 and Sec. B.5.1]. The mask itself is anchored to the -40 dBc bandwidth B_{-40} of the pulse [12, Sec. B.4.2]. From the point at which the emission falls to -40 dBc, the mask rolls off at 30 dB per decade until it meets the spurious emission limit [12, Sec. B.2]. That limit is the least stringent of $43+10\log_{10}(\text{PEP})$ or 60 dB below carrier power, which for most marine radars amounts to -60 dBc [12, Sec. B.2]. Where B_{-40} falls within the allocated band the mask commences at the edge of that band, whereas if it falls outside the mask commences at that point in the adjacent band [12, Fig. B.1 and Fig. B.2]. Compliance requires the measured emission spectrum to remain below this mask [12, Sec. B.5.3], confirmed

by measurement and analytical evaluation [12, Sec. 6.2.1.2]. IEC are interested in the compliance of these requirements. Power amplifier nonlinear effects broaden the transmitted spectrum through spectral regrowth and harmonic generation, which risks emissions rising above the mask – violating the requirement. This thesis work may present solutions to countermand these effects, ensuring spectral compliance with the standard.

- **Radar System Users:** Users of the radar system, namely crew members responsible for navigating their vessel at sea, are interested in the results of this thesis, because it will potentially improve performance and reliability of the radar image used for navigating safely. The solutions presented in this thesis may aid in reducing masking around strong targets – enabling weaker targets within their vicinity to be distinctly detected. There is also the risk of the emergence of false targets – which this work may potentially mitigate. This helps crews to be sure that they can trust their navigation instruments. Performance gains, such as increased range and detection of smaller targets, may also help crews navigate more safely to avoid collision.
- **The Student:** The student, the author of this thesis, is interested in learning about the technicalities of radar development and deepening their understanding of concepts in signal processing. This thesis work provides the student the opportunity to gain practical experience in these aspects.

3.4 Objectives Analysis

The overarching objective of this work is to determine what real effects power amplifier nonlinearity has on the pulse compression radar system and appropriate predistortion techniques to amend such effects for performance gain and spectral compliance. To better solidify how this can be achieved, it is divided into several detailed objectives below.

3.4.1 Generic Modelling of the SSR-X Pulse Compression Radar

The first objective is to develop a pulse compression radar simulation that can be used as a controlled platform for nonlinearity and digital predistortion experiments. This

simulation serves as a generic model to understand the general behaviour of the power amplifier chain and apply different parameter values and view how this affects the outputs. Within such a simulation, various power amplifier nonlinearity models from literature can be dynamically applied and their effects investigated. On that note, a simulation will also demonstrate what the outputs of an ideal system would be, void of nonlinear effects. Furthermore, the simulation forms the basis for experimenting with different digital predistortion algorithms.

To best understand the power amplifier chain behaviour, it stands to reason for the simulation to simulate the main stages of the pulse compression radar and pulses generated by the real system in its standard operating mode. The pulses used by the pulse compression radar comprise a medium and long pulse – chosen depending on desired range. These pulses are generated in the real system's FPGA as baseband signals. They are then windowed, upconverted to IF, converted to analogue signals, then upconverted to RF preceding amplification for transmission. This forms the transmit chain. Each pulse would return an attenuated and time delayed echo then passed through the receive chain of the system. In the receive chain, the echoes are amplified, down converted back to IF and baseband frequencies, and low pass filtered at the IF. The final stage of the chain entails pulse compression to maximise SNR and detect after which delay time the echo returns. Each of the above stages act as primary components forming the pulse compression radar system.

Furthermore, the simulation should produce plots that can be used to quantitively analyse power amplifier nonlinearity effects. Plots relevant to this topic are namely spectra of the various signals across the chain, AM-AM and AM-PM curves of the power amplifier, and the pulse compression output. Therefore, it stands to reason for the model to generate said plots. To evaluate overall performance, relevant plots should include quantitative values for KPIs pertinent to known nonlinearity effects – these primarily encompass signal magnitude, side lobe level, harmonics level, AM-AM linearisation, and AM-PM phase deviation. To elaborate, pulse compression and spectrum plots should measure main lobe peak amplitude, PSL, and ISL as metrics to determine effects on signal magnitude, spectral compliance and range side lobes. Spectrum plots should additionally measure main lobe level of the second harmonic to determine extent of harmonics' effects. AM-AM curves should measurably highlight the 1 dB compression point as a KPI for linearisation effect, whereas total phase deviation should constitute the KPI for AM-PM curves.

This objective is achieved if the simulation is developed and supports both pulses and the experimentation of power amplifier nonlinearity and digital predistortion models – involving the generation of relevant simulation plots with quantitative values for KPIs.

3.4.2 Precise Modelling of the SSR-X Pulse Compression Radar

Following completion of the generic pulse compression radar model, the second objective is to adjust the parameters of the simulation to best model the actual pulse compression radar hardware. This will allow experiments on the nonlinearities and digital predistortion to be relevant to the real-world system. This will result in understanding the actual effects nonlinearity has on the pulse compression radar system hardware and how it reduces performance.

This objective involves performing lab measurements on the pulse compression radar hardware to obtain an understanding of real system behaviour. The measurement step should endeavour to produce any relevant plots that help determine accurate parameters and settings for the simulation. The plots that can be realistically derived from measurement are dependent on the confines and available connectors of the hardware. Hence, it is unrealistic to obtain all the same plots generated by the simulation.

When sufficient measurement plots and values are obtained, simulation parameters can be finetuned to emulate the hardware as closely as possible. The simulation should thus ideally include a variety of power amplifier nonlinearity models, so a comparison can be made between models to determine which best matches the real system. This is conducive to understanding the actual nonlinearity effects on the hardware and deducing its effects on performance.

This objective is achieved when adequate measurements for behavioural understanding are obtained and the simulation finetuned with the best matching power amplifier model chosen to precisely emulate the power amplifier chain hardware.

3.4.3 Implementation of Digital Predistortion

Once concrete nonlinear effects are determined from the precise model, the final objective involves implementing digital predistortion to counteract these effects and improve

performance. This would involve the application of digital predistortion algorithms in the simulation, where the counter-effects on nonlinearity can be observed in the generated simulation results. To compare and evaluate results, it stands to reason to generate three separate sets of results – one ideal, one with nonlinear effects modelled, and one with digital predistortion applied. Values of the curves' KPIs – signal magnitude, side lobe level, harmonics level, AM-AM nonlinear deviation, and AM-PM phase deviation – can then be directly compared. This aids in providing a metric to quantitively evaluate the extent of performance gain digital predistortion introduces. Time permitting, this same comparison and evaluation could potentially also be performed on the real pulse compression radar hardware by implementing the simulated digital predistortion algorithm in its firmware.

This objective is achieved when a digital predistortion algorithm is implemented in the simulation (and possibly in the firmware) and ideal, nonlinear, and predistorted results are compared with subsequent performance gain evaluated.

3.5 Requirements Overview

Based on the system environment and analysis of stakeholders and objectives, the following functional and non-functional requirements are derived. Functional requirements describe behaviours that the system must provide, whereas non-functional requirements describe that quality constraints the system must adhere to e. g. performance, reliability and usability. These requirements must be concrete and testable to determine whether the developed system has truly fulfilled them. The requirements directly impact decisions on system design and final evaluation.

3.5.1 Functional Requirements

FR1: The pulse compression radar simulation shall include the following modular stages for complete modelling of the real system:

- Baseband signal generation
- Baseband signal windowing
- Up conversion to IF

- Up conversion to RF
- Transmit chain amplification (power amplifier)
- Channel effects (delay and attenuation)
- Receive chain amplification
- Down conversion to IF
- IF low pass filter
- Down conversion to Baseband frequency
- Pulse compression

FR2: The pulse compression radar simulation shall include the medium (M1) and long (L1) pulses generated by the real system respectively.

FR3: The pulse compression radar simulation shall apply at least two known power amplifier nonlinearity models for comparison.

FR4: The pulse compression radar simulation shall apply at least one known digital predistortion algorithm for evaluation.

FR5: For each transmit pulse type (medium and long), the pulse compression radar simulation shall generate analysis plots that overlay the ideal, nonlinear power amplifier, and predistorted power amplifier cases for direct comparison.

FR6: Generated pulse compression radar simulation spectrum plots shall include quantitative values for the following KPIs for evaluation:

- Main lobe peak amplitude
- PSL
- ISL
- Second harmonic main lobe level

FR7: Generated pulse compression radar simulation pulse compression plots shall include quantitative values for the following KPIs for evaluation:

- Main lobe peak amplitude

- PSL
- ISL

FR8: Generated pulse compression radar simulation AM-AM plots shall include quantitative values for the KPI of 1 dB compression point for evaluation.

FR9: Generated pulse compression radar simulation AM-PM plots shall include quantitative values for the KPI of total phase deviation for evaluation.

3.5.2 Non-Functional Requirements

NFR1: The pulse compression radar simulation shall be developed in Python to ease the learning curve and development speed for the student due to its simple built-in libraries and the student's familiarity with it.

NFR2: The pulse compression radar simulation shall be developed modularly, so that it can be easily maintained and extended.

NFR3: The documentation and implementation of the work shall be well structured and comprehensible to ensure future developers or researchers can inspect, reproduce, or extend the work. This entails clear project structure, fully documented project components, and repletely annotated results.

4 Concept and Design

This chapter presents the concept and design of the investigative work of this thesis. The previous chapter outlined research direction, requirements, and objectives of the system to be developed. Based on these requirements, this chapter discusses the design of the system preceeding implementation in Chapter 5.

The main purpose of the design is to develop a suitable SSR-X pulse compression radar simulation and accompanying evaluation framework to facilitate the investigation of power amplifier nonlinearity effects and predistortion to counteract these. As elaborated in the objectives analysis section 3.4, this involves first developing a generic pulse compression radar simulation – modelling the main components of the real SSR-X pulse compression radar system, as well as nonlinear power amplifier and predistorter models. Then, the simulation is tailored to accurately emulate the real system using measurements performed on the physical hardware. The simulation serves as a boilerplate framework to apply and test predistortion algorithms preceeding any implementation on the real system. As such, evaluation criteria are necessary for judgement of the performance of predistortion. This involves numerical comparison predicated on the different KPIs mentioned in Chapter 3.

The design of the components of the system are of a modular structure in accordance with NFR2 in in section 3.5. This ensures an easily maintable and extendable system. Each component of the pulse compression chain, the power amplifier, and predistorter model are developed independently from each other as modules within the simulation. They interact together, but can be modified and finetuned separately. As such, observed performance differences between different power amplifier and predistorter models for comparison can be strictly attributed to changes in those modules, not to other simulation conditions. This enables a reliable framework for evaluation.

4.1 Overall Approach

The overall approach of this thesis is to design and implement the SSR-X pulse compression radar simulation and use it as a test environment for evaluating and understanding the effects of power amplifier nonlinearity and predistortion preceeding hardware implementation. Development of the simulation is the core of the work, enabling a test sandbox to easily configure and experiment with nonlinearity effects and predistortion results. Through refinement by measurements on the hardware, the true behaviour of the system can be closely modelled. Once the best predistortion technique is found in simulation, the experiment can be conducted on the hardware to observe for real world performance gains.

The general workflow of this thesis is shown in the following steps:

1. Design and implement a generic SSR-X pulse compression radar simulation – including main components of the pulse compression system, the power amplifier, and predistorter
2. Perform measurements on the SSR-X pulse compression radar hardware to better understand the behaviour of the real system
3. Finetune the simulation using measurement observations and findings for accurate representation of the system
4. Apply evaluation criteria to determine the performance improvement by predistortion in simulation
5. Apply the simulated predistorter on the SSR-X pulse compression radar hardware to confirm real performance improvement

The simulation and its accompanying evaluative aspects are implemented as a python script, in accordance with NFR1 in section 3.5. This is conducive to rapid development and implementation ease due to the abundance of useful provided libraries. Real system hardware measurements however will be performed in a lab environment using measurement equipment, such as oscilloscopes and spectrum analysers.

The application of evaluation criteria for predistorter performance is based on numerically comparing KPI metrics listed in section 3.5. In the simulation, this evaluation will dictate how much predistortion theoretically enhances performance by cancelling

adverse nonlinearities. When the predistorter is then implemented on the physcial hardware, the evaluation can again be performed using measured KPIs – verifying performance improvement in practicality.

4.2 System Architecture

The system architecture is designed as a modular SSR-X pulse compression radar simulation system and evaluation framework. The primary components entail the pulse compression radar system, nonlinear power amplifier, predistorter, and evaluative result logs and plots. Each of these primary components, including their relationship with one another, is elaborated on in the following subsections.

4.2.1 Pulse Compression Radar Chain Design

The pulse compression radar system serves as the backbone of the entire simulation. It comprises each of the main building blocks of the pulse compression system, namely the transmit chain, channel, and receive chain. Each of these building blocks consists of further subcomponents. In the transmit chain, the submodules are baseband signal generation and Kaiser windowing, predistortion (explained in section 4.2.3), upconversion to IF and RF respectively, amplification (initially via an ideal linear amplifier, though extended in section 4.2.2), and harmonic filter before transmission. In the receive chain, the submodules are the LNA, harmonic filter, downconversion to IF, IF low pass filter, downconversion to baseband, and finally pulse compression. The modules have been listed in chronological order where the output of one module becomes the input of the next.

Traversing each of these submodules are the radar pulse signals themselves. SSR-X pulse compression radar signals of interest for the thesis' investigation are the LFM medium (M1) and long (L1) pulses, where the M1 pulse has a 4 µs duration and 10 MHz bandwidth and the L1 pulse a 25 µs duration and 5 MHz bandwidth.

The pulses are first generated in the baseband signal generation module and must be passed and processed through the entire signal chain until the pulse compression submodule. The submodules are thus designed as modular Python functions, and the pulses are sequentially processed as Python objects by each function – in accordance

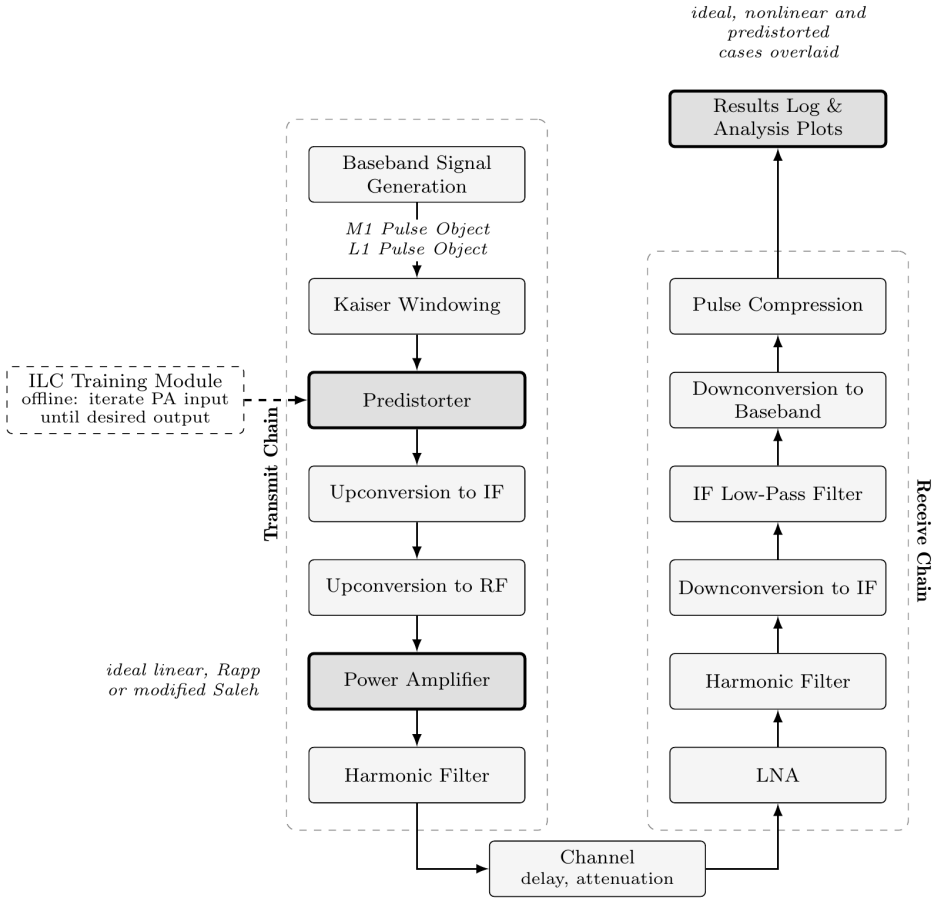


Figure 4.1: Programme flow of pulse compression radar simulation

with FR2 in section 3.5. A complete programme flow diagram of the design is seen in Figure 4.1.

4.2.2 Power Amplifier Module Design

The power amplifier module models the nonlinearity effects of the power amplifier chain of the system. It essentially stands as the (nonlinear) amplification module in the transmit chain of the pulse compression radar simulation – in essence, the module is designed as Python function and is depicted in Figure 4.1. Section 2.3 elucidates various proven power amplifier mathematical models from theory. The function applies

the mathematical model on the input pulse signal to obtain the distorted output pulse afflicted with nonlinearity effects.

Due to its simplicity and reliability when extrapolating towards saturation, the Rapp model – along with Honkanen and Häggman's AM-PM extension – is chosen as the main mathematical model. Details of the Rapp model are found in section 2.3.3. As a secondary model, the modified Saleh model – introduced by O'Droma, Meza and Lei and seen in section 2.3.3 – is chosen due to it being the best performing model when extending into saturation. The two models can be compared against measurements performed on the real hardware to find the most suitable option. The usage of these two models satisfies FR3 in section 3.5.

4.2.3 Predistorter Module Design

Analogous to the power amplifier module, the predistorter module is a Python function serving as a submodule in the transmit chain – seen in Figure 4.1. It is inserted directly after baseband signal generation and windowing. Section 2.4 introduces three types of predistortion – baseband, IF, and RF predistortion. The predistorter module design will focus on digital baseband predistortion due to its simplicity, reconfigurability, and ease of implementing adaptivity. Three baseband predistortion approaches are discussed in section 2.4.2 – ILA, DLA, and ILC. Of the three, ILC is the most suitable approach for pulse compression radar. As the radar pulses remain the same for every transmission – the M1 and L1 pulses respectively – the optimal predistorter result can be computed once and saved. No model of the power amplifier itself nor its inverse characteristic are required like the ILA and DLA approaches. However, the design is only made adaptive by periodically recomputing the optimal predistorter result via ILC. This only concerns implementation of the predistorter module on real hardware, as the simulation is unaffected by the factors that adaptivity aims to counter i. e. aging and temperature effects.

Hence, ILC is chosen for the predistorter module design, fulfilling FR4 in section 3.5. The predistorter module function outputs the precalculated optimal predistorted pulse depending on its input, either the M1 or L1 pulse. The optimal predistorter result is computed by running a training module preceeding full programme flow through the pulse compression radar simulation. The training module iteratively runs the ILC algorithm by passing the input pulse through the power amplifier module and adjusting

it until the desired output is achieved. Once the desired output is obtained, it is saved as the predistorter output and loaded when the predistorter module is called during a full programme run.

4.2.4 Results Logs and Plots

The plotting and results logs are critical components for the determination of nonlinearity effects and performance gain from predistortion. The requirements in section 3.5 define the required plots and recording of KPIs for this purpose. FR5 states that analysis plots are required for three cases – ideal, nonlinear, and predistorted – overlayed on each other for direct visual comparison. FR6, FR7, FR8, and FR9 define the plots to be generated, including quantitative values of KPIs the plots should display. The generated plots, including their KPIs, entail the following:

1. Spectrum plots at each stage of the simulation with KPIs of main lobe peak amplitude, PSL, ISL, and second harmonic main lobe level
2. Pulse compression plot of with KPIs of main lobe peak amplitude, PSL, and ISL
3. Power amplifier AM-AM plots with KPI of 1 dB compression point
4. Power amplifier AM-PM plots with KPI of total phase deviation

As stated, the above analysis plots will overlay the ideal, nonlinear, and predistorted curves – including markers for their defined KPIs. The results logs will print the KPI numerical values and their differences for the three cases in a log file for evaluation. The results log and plot generation are respectively designed as python functions. These functions are called at the end of the simulation, following execution of all stages of the pulse compression radar simulation. They are seen as the final stage depicted in Figure 4.1.

4.3 Evaluation Criteria

Evaluation here is two-fold. Firstly, evaluation refers to determining the extent of detrimental nonlinearity and performance gain via predistortion. Secondarily, evaluation also refers to determining whether the developed simulation and thesis work are

satisfactory – in fulfillment with the requirements defined in section 3.5. This requirements fulfillment evaluation is performed in Chapter 6.

Regarding the nonlinearity extent and performance gain, evaluation is predicated on the generated results logs and plots elaborated in section 4.2.4. The evaluation is purely investigative and numerically grounded. The extent of power amplifier nonlinearity will be quantitatively determined by comparing differences in KPI values between the ideal (linear) and nonlinear case. On the other hand, the extent of predistortion performance gain is quantitively deduced by comparing differences in between nonlinear and predistorted cases – to examine the extent of nonlinearity reversal – and between linear and predistorted cases – to examine how close predistortion can reach the ideal linearity. The evaluation will be performed in Chapter 6 in tabular form to clearly see numerical discrepancies between the three cases. The descriptive evaluation obtained from these results will follow.

5 Implementation

This chapter presents the implementation of the investigative work of this thesis. This involves implementation of the SSR-X pulse compression radar simulation and its accompanying evaluation framework to investigate power amplifier nonlinearity effects and predistortion benefits. The previous chapter outlined the architecture of the simulation and evaluation system to be developed, including the general workflow approach for implementation. The implementation follows this workflow and develops the system and its components based on the design details of Chapter 4.

The following subsections first discuss the implementation of the generic SSR-X pulse compression radar simulation, its components, and its evaluative framework. Then, the refinement of the simulation through hardware measurements is explored. Finally, implementation of the derived simulated predistorter on real hardware is outlined. Unforeseen challenges and modifications that became apparent during implementation are also mentioned. The final chapters 6 and 7 report on the results and drawn conclusions from the experiments conducted during implementation.

The aim of this chapter is to describe implementation of the work in detail to provide a solid understanding of the developed system and experimentive work – without the need to inspect the source code, experimental setup, or measurement execution directly.

5.1 Development of Generic SSR-X Pulse Compression Radar Simulation

The generic SSR-X pulse compression radar simulation is implemented as Python software, simulating the primary components of the SSR-X pulse compression radar and generating apposite plots and results logs for analysis. All of these aspects of the simulation are implemented as independent modular functions that interact with one another to emulate the complete pulse compression radar chain – described more thoroughly in

subsection 4.2. This section details the implementation of each of these aspects of the simulation.

The generic simulation lays the foundational scaffolding for nonlinearity analysis and determining predistorter performance enhancement. As such, it is implemented as a general draft representation of the real SSR-X pulse compression radar hardware – only considering primary components and general behaviour of pulse compression radar. The correct numerical values provided to and produced by each component to best model real system behaviour is later determined by real system measurements to finetune the generic simulation.

As mentioned, the simulation is developed in Python in accordance with NFR1 in section 3.5. Python was selected, because it is replete with multifarious ready-made library modules appropriate for simulation, calculating, and plotting. Moreover, the author of the thesis is more familiar with the language, ameliorating the learning curve. The following subsections, describing each simulation aspect's implementation, mentions the specific library modules utilised.

5.1.1 SSR-X Pulse Compression Radar Chain

The SSR-X pulse compression radar chain is the structural backbone of the entire simulation. It comprises a chain of sequentially cascading subcomponents in a transmit, channel, and receive path that form the basis of a pulse compression system.

5.1.1.1 Pulse Signal Class

5.1.1.2 Kaiser Window Module

5.1.1.3 IF and RF Mixing Module

5.1.1.4 Power Amplifier Module

SAY NO HARMONIC FILTER

5.1.1.5 Channel Module

5.1.1.6 LNA Module

5.1.1.7 IF Low Pass Filter Module

5.1.1.8 Pulse Compression Module

The power amplifier module serves as the heart of the simulation. It is here that nonlinearity is generated in the simulation.

5.1.2 Results Logs and Plotting

5.2 Development of Precise SSR-X Pulse Compression Radar Simulation

5.2.1 Measurement of SSR-X Pulse Compression Radar Hardware

5.2.2 Refinement of SSR-X Pulse Compression Radar Simulation

5.3 Development of Predistortion on SSR-X Pulse Compression Radar Hardware

6 Evaluation

7 Conclusion and Outlook

Bibliography

[1] Abi Hussein, M. ; Bohara, V. A. ; Venard, O.: On the System Level Convergence of ILA and DLA for Digital Predistortion. In: *Proceedings of the International Symposium on Wireless Communication Systems (ISWCS)*. Paris, France : IEEE, 2012, S. 870–874

[2] Chani-Cahuana, J. ; Landin, P. N. ; Fager, C. ; Eriksson, T.: Iterative Learning Control for RF Power Amplifier Linearization. In: *IEEE Transactions on Microwave Theory and Techniques* 64 (2016), September, Nr. 9, S. 2778–2789. http://dx.doi.org/10.1109/TMTT.2016.2588483. – DOI 10.1109/TMTT.2016.2588483

[3] Cottais, E. ; Wang, Y. ; Toutain, S.: A New Adaptive Baseband Digital Predistortion Technique. In: *European Microwave Association* 2 (2006), S. 154–159

[4] Ding, L. ; Zhou, G. T. ; Morgan, D. R. ; Ma, Z. ; Kenney, J. S. ; Kim, J. ; Giardina, C. R.: A Robust Digital Baseband Predistorter Constructed Using Memory Polynomials. In: *IEEE Transactions on Communications* 52 (2004), Nr. 1, S. 159–165

[5] Durukal, Mustafa: *Power Amplifier Linearization by Predistortion*. Ankara, Turkey, Bilkent University, Master's thesis, September 2006

[6] Eun, C. ; Powers, E. J.: A New Volterra Predistorter Based on the Indirect Learning Architecture. In: *IEEE Transactions on Signal Processing* 45 (1997), Nr. 1, S. 223–227

[7] Feng, Xiaowen ; Wang, Yide ; Feuvrie, Bruno ; Descamps, Anne-Sophie ; Ding, Yuehua ; Yu, Zhiwen: Analysis on LUT Based Digital Predistortion Using Direct Learning Architecture for Linearizing Power Amplifiers. In: *EURASIP Journal on Wireless Communications and Networking* 2016 (2016), Nr. 1, S. 132.

http://dx.doi.org/10.1186/s13638-016-0628-y. – DOI 10.1186/s13638–016–0628–y

[8] Fisher, Paul O. ; Al-Sarawi, Said F.: Improving the Accuracy of SSPA Device Behavioral Modeling. In: *Proceedings of the 2015 International Conference on Information and Communication Technology Research (ICTRC)* IEEE, 2015. – ISBN 9781479989669, S. 275–278

[9] Ghorbani, A. ; Sheikhan, M.: The Effect of Solid State Power Amplifier (SSPA) Nonlinearities on MPSK and M-QAM Signal Transmission. In: *Proceedings of the 6th International Conference on Digital Processing of Signals in Communications, Loughborough, UK*, 1991, S. 193–197

[10] Hanning, Lowisa: *Pulse Shaping of Radar Transmitters: Compensation of Memory Effects through Digital Pre-distortion.* Gothenburg, Sweden, Chalmers University of Technology, Master's thesis, 2018

[11] Honkanen, M. ; Häggman, S.-G.: New Aspects on Nonlinear Power Amplifier Modeling in Radio Communication System Simulations. In: *Proceedings of the 8th IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), Helsinki, Finland* IEEE, 1997, S. 844–848

[12] International Electrotechnical Commission (Hrsg.): *IEC 62388: Maritime navigation and radiocommunication equipment and systems – Shipborne radar – Performance requirements, methods of testing and required test results.* Edition 2.0. Geneva: International Electrotechnical Commission, 2013

[13] Kenington, Peter B.: *High Linearity RF Amplifier Design.* Norwood, MA : Artech House, 2000. – ISBN 9781580531436

[14] Kim, J. ; Konstantinou, K.: Digital Predistortion of Wideband Signals Based on Power Amplifier Model with Memory. In: *Electronics Letters* 37 (2001), Nr. 23, S. 1417–1418

[15] Lee, Thomas H.: *The Design of CMOS Radio-Frequency Integrated Circuits.* Cambridge, UK : Cambridge University Press, 1998. – ISBN 9780521639224

[16] Li, F. ; Feuvrie, B. ; Wang, Y. ; Chen, W.: MP/LUT Baseband Digital Predistorter for Wideband Linearisation. In: *Electronics Letters* 47 (2011), Nr. 19, S. 1096–1098

[17] Li, H. ; Kwon, D. H. ; Chen, D. ; Chiu, Y.: A Fast Digital Predistortion Algorithm for Radio-Frequency Power Amplifier Linearization with Loop Delay Compensation. In: *IEEE Journal of Selected Topics in Signal Processing* 3 (2009), Nr. 3, S. 374–383

[18] Maas, Stephen A.: *Nonlinear Microwave and RF Circuits*. 2nd edition. Norwood, MA : Artech House, 2003. – ISBN 9781580534840

[19] Mar, Thin T. ; Mon, Su Su Y.: Pulse Compression Method for Radar Signal Processing. In: *International Journal of Science and Engineering Applications* 3 (2014), Nr. 2, S. 31–35. – ISSN 2319–7560

[20] Maršálek, Roman: *Power Amplifier Linearization Using Digital Baseband Adaptive Predistortion*. Marne-la-Vallée, France, Université de Marne-la-Vallée, PhD thesis, 2003

[21] Miehle, Konrad: *A New Linearization Method for Cancellation of Third Order Distortion*. Charlotte, NC, University of North Carolina at Charlotte, Master's thesis, 2003

[22] Mkadem, F. ; Boumaiza, S.: Physically Inspired Neural Network Model for RF Power Amplifier Behavioral Modeling and Digital Predistortion. In: *IEEE Transactions on Microwave Theory and Techniques* 59 (2011), Nr. 4, S. 913–923

[23] Morgan, D. R. ; Ma, Z. ; Ding, L.: Reducing Measurement Noise Effects in Digital Predistortion of RF Power Amplifiers. In: *Proceedings of the IEEE International Conference on Communications (ICC)* Bd. 4, IEEE, 2003, S. 2436–2439

[24] Morgan, D. R. ; Ma, Z. X. ; Kim, J. ; Zierdt, M. G. ; Pastalan, J.: A Generalized Memory Polynomial Model for Digital Predistortion of RF Power Amplifiers. In: *IEEE Transactions on Signal Processing* 54 (2006), Nr. 10, S. 3852–3860

[25] Naskas, N. ; Papananos, Y.: Neural-Network-Based Adaptive Baseband Predistortion Method for RF Power Amplifiers. In: *IEEE Transactions on Circuits and Systems II: Express Briefs* 51 (2004), Nr. 11, S. 619–623

[26] O'Droma, M. ; Meza, S. ; Lei, Y.: New Modified Saleh Models for Memoryless Nonlinear Power Amplifier Behavioural Modelling. In: *IEEE Communications Letters* 13 (2009), June, Nr. 6, S. 399–401

[27] Pozar, David M.: *Microwave Engineering*. 4th edition. Hoboken, NJ : John Wiley & Sons, 2012. – ISBN 9780470631553

[28] Rapp, C.: Effects of HPA-Nonlinearity on a 4-DPSK/OFDM-Signal for a Digital Sound Broadcasting System. In: *Proceedings of the 2nd European Conference on Satellite Communications, Liège, Belgium*, 1991, S. 179–184

[29] Razavi, Behzad: *RF Microelectronics*. 2nd edition. Upper Saddle River, NJ : Prentice Hall, 2011 (Prentice Hall Communications Engineering and Emerging Technologies). – ISBN 9780137134731

[30] Richards, Mark A. (Hrsg.) ; Scheer, James A. (Hrsg.) ; Holm, William A. (Hrsg.): *Principles of Modern Radar: Volume I – Basic Principles*. Raleigh, NC : SciTech Publishing, 2010. – ISBN 9781891121524

[31] Saleh, Adel A. M.: Frequency-Independent and Frequency-Dependent Nonlinear Models of TWT Amplifiers. In: *IEEE Transactions on Communications* 29 (1981), November, Nr. 11, S. 1715–1720

[32] Schreurs, Dominique ; O'Droma, Máirtín ; Goacher, Anthony A. ; Gadringer, Michael: *RF Power Amplifier Behavioral Modeling*. Cambridge, UK : Cambridge University Press, 2009 (The Cambridge RF and Microwave Engineering Series). – ISBN 9780521881739

[33] Shi, B. ; Sundstrom, L.: A 200-MHz IF BiCMOS signal component separator for linear LINC transmitters. In: *IEEE Journal of Solid-State Circuits* 35 (2000), July, Nr. 7, S. 987–993. http://dx.doi.org/10.1109/4.848207. – DOI 10.1109/4.848207

[34] Singla, R. ; Sharma, S.: Digital Predistortion of Power Amplifiers Using Look-Up Table Method with Memory Effects for LTE Wireless Systems. In: *EURASIP Journal on Wireless Communications and Networking* 2012 (2012), Nr. 1, S. 330

[35] Singla, R. ; Sharma, S.: Low Complexity Look Up Table Based Adaptive Digital Predistorter with Low Memory Requirements. In: *EURASIP Journal on Wireless Communications and Networking* 2012 (2012), Nr. 1, S. 43

[36] Skolnik, Merrill I. (Hrsg.): *Radar Handbook*. 3rd edition. New York, NY : McGraw-Hill, 2008. – ISBN 9780071485470

[37] Wambacq, Piet ; Sansen, Willy M. C.: *The Kluwer International Series in Engineering and Computer Science*. Bd. 451: *Distortion Analysis of Analog Integrated Circuits*. Boston, MA : Kluwer Academic Publishers, 1998. – ISBN 9780792381860

[38] White, G. P. ; Burr, A. G. ; Javornik, T.: Modelling of Nonlinear Distortion in Broadband Fixed Wireless Access Systems. In: *Electronics Letters* 39 (2003), April, S. 686–687

[39] Zhou, D. ; DeBrunner, V. E.: Novel Adaptive Nonlinear Predistorters Based on the Direct Learning Algorithm. In: *IEEE Transactions on Signal Processing* 55 (2007), Nr. 1, S. 120–133

A Appendix

A.1 Test Series

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Declaration on the Use of AI Tools

I hereby declare that I have used no other AI-based tools and for no other purpose than detailed in the table below.

AI tool	**Used for**	**Chapters**	**Remarks**
ChatGPT (GPT-5)	Support with literature research: suggestions for German and English search terms regarding modeling approaches and numerical solution methods for the system under investigation, as well as a summary of abstracts for the preliminary selection of relevant publications.	2.1–2.3	Accessed in March 2026. The actual research was conducted using specialized databases (e. g., IEEE Xplore, ScienceDirect). All cited sources were read in their original form; the AI-generated summaries served only for preliminary selection and are not used as sources.
Claude (Claude Sonnet 5.5)	Code generation and debugging assistance for Python helper functions: reading and processing measurement and reference data (pandas), plotting functions for simulation results (matplotlib), and troubleshooting an incorrect unit conversion during data processing.	4.2, App. B	The model equations, the implementation of the simulation model, and the evaluation logic were developed without AI assistance. The generated code was reviewed, validated using test datasets, and marked with comments in the source code.
ChatGPT (GPT-5)	Linguistic revision of one's own text drafts: suggestions regarding style and sentence structure, as well as translation of the German summary into English (abstract).	1 and 6, abstract	Only original texts containing no personal or confidential data were entered. Suggestions were reviewed individually and adopted only in part; technical statements and terminology were not altered by the AI.

Declaration of Authorship

I hereby declare that I have elaborated this thesis, including all its parts, using no other sources, references, or aids than specified in this work. I have disclosed all AI-based tools used, including product names and version numbers, where applicable.

I take over the full responsibility for all machine-generated content; and I am fully liable for erroneous or misleading content, incorrect references, violations of data protection or copyright laws, or instances of plagiarism resulting from the use of AI.

City	Date	Original signature